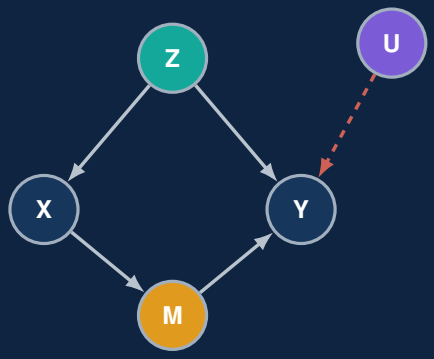


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 11

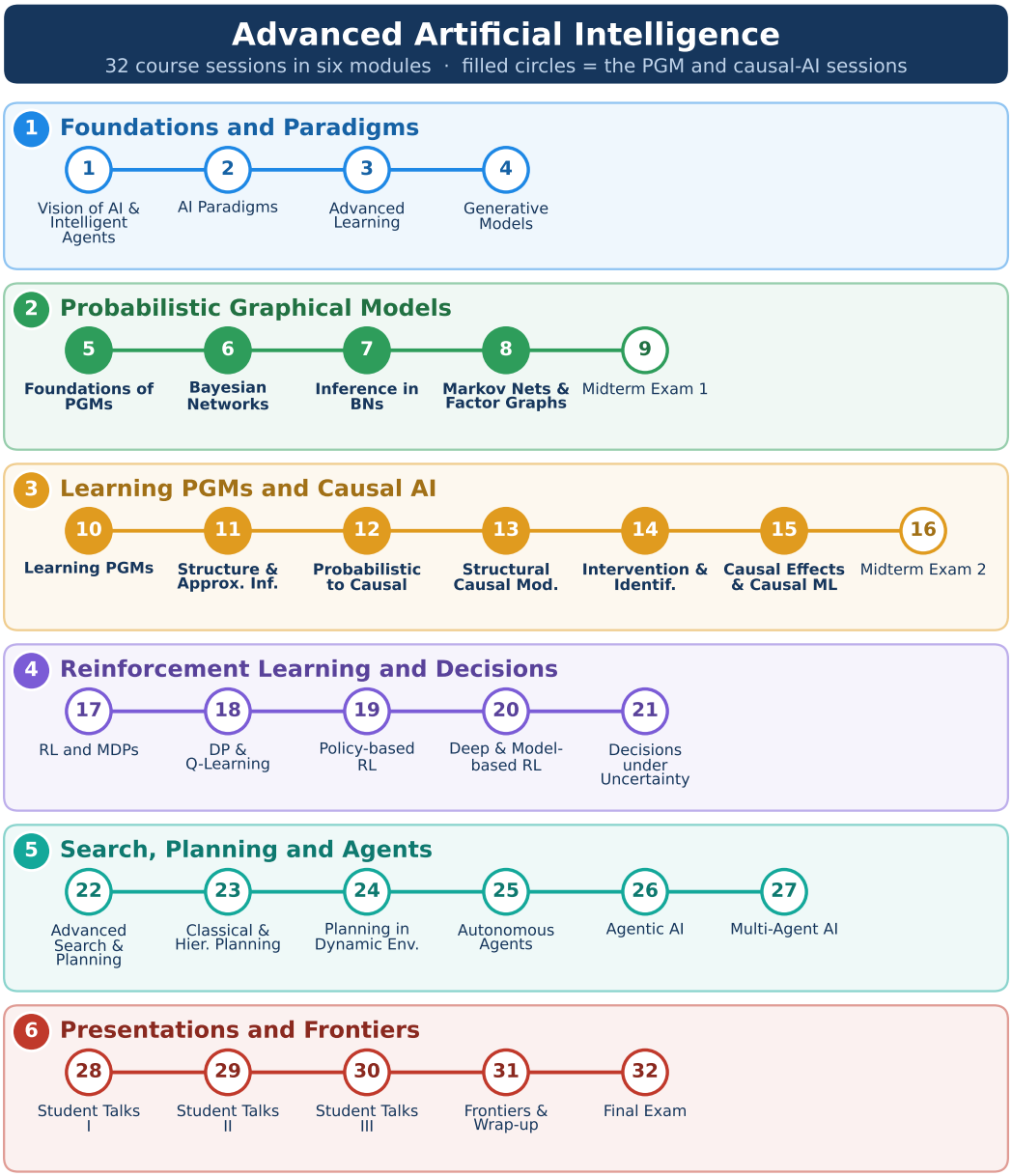
Structure Learning and Approximate Inference

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 11 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s11_examples.py` for the numbers, `python/s11_figures.py` for the figures), and `questions/s11_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

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SESSION 11 ADVANCED ARTIFICIAL INTELLIGENCE

Structure Learning and Approximate Inference

Model selection, Monte Carlo sampling, MCMC and approximate inference

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Structure Learning
Scores, model selection, greedy search, the PC algorithm and CPDAGs.

PART II
Monte Carlo
Ancestral, rejection and importance sampling, likelihood weighting, ESS.

PART III
MCMC and Beyond
Gibbs, Metropolis–Hastings, diagnostics, variational inference, particle filters.

PART IV
Mathematical Statistics
Monte Carlo CLT, optimal proposals, Markov-chain CLT, BIC asymptotics.

PART V
AI Connections
MCTS, belief tracking, self-consistency, world models, agent graphs.

LEARNING OBJECTIVES

- Explain why likelihood prefers the complete DAG; compute AIC, BIC, Bayesian (BDe/K2) and held-out scores, search by hill climbing, and run the PC algorithm to a CPDAG.
- Build Monte Carlo estimators (ancestral, rejection, likelihood weighting, importance sampling), quantify their error with the CLT, and diagnose weight degeneracy with the effective sample size.
- Derive the Metropolis–Hastings acceptance ratio, prove that detailed balance implies stationarity, and read MCMC output critically (burn-in, autocorrelation, ESS, $\hat{R}$).
- Set up mean-field VI and a particle filter, and explain the asymmetry of the KL divergence.
- Connect sampling and structure learning to MCTS, belief tracking, LLM self-consistency, world models and multi-agent graphs.

Where we are. Sessions 5–8 (course sessions 5–8) built Bayesian and Markov networks and exact inference; Session 10 learned the *parameters* of a fixed graph. Today we learn the *graph* itself, and replace exact inference by sampling and optimisation when treewidth makes it hopeless, closing the probabilistic half of the notes. Session 12 opens the causal half by asking what a learned graph can and cannot tell us about interventions; sampling returns in RL (course sessions 17–20), planning (course session 22) and belief tracking (course session 21).

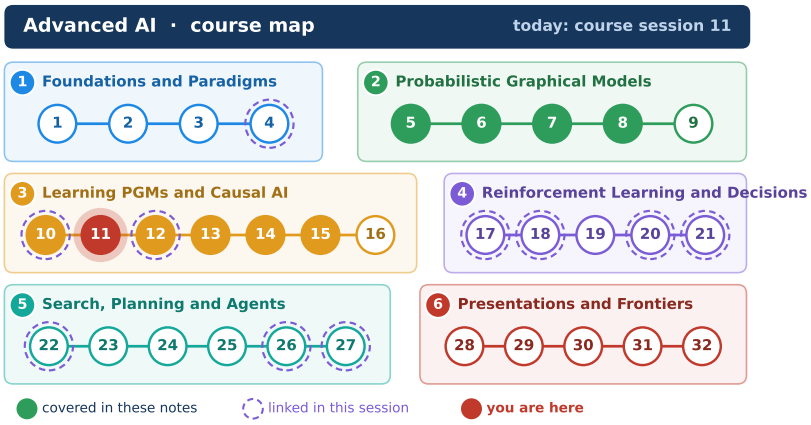


Figure 11.1. Where this session sits in the Advanced AI course (course session 11). Dashed purple rings mark the sessions this chapter links to explicitly.

OPENING MOTIVATION

Two loose threads remain from Session 10. First, every session so far assumed the graph itself — who is a parent of whom, which pairs of variables share an edge — was simply handed to us by a domain expert. What if it isn't? Second, the EM algorithm's E-step leaned entirely on Session 7's variable elimination, but VE's cost is exponential in treewidth — what happens in the many realistic models (dense social networks, image grids, large medical ontologies) where treewidth is too large for exact inference to run in any reasonable time? This session closes both threads: briefly, for structure; at length, via sampling, for inference. The sampling ideas developed here — Monte Carlo error analysis and, above all, importance weighting — will return in Session 15, where inverse propensity weighting turns out to be importance sampling and Monte Carlo/bootstrap error bars quantify the uncertainty of causal effect estimates.

Healthcare

Hospitals learn a comorbidity graph from records, then query it by Gibbs sampling because it is too dense for exact elimination.

Robotics

A robot localises itself with a particle filter: a cloud of weighted hypotheses that is moved, re-weighted by each sensor reading and resampled.

LLM agents

An agent samples several reasoning paths and votes (self-consistency), or rolls out plans in a search tree: Monte Carlo estimation with a language model as the sampler.

PART I

Learning the Graph from Data

Scores that resist overfitting, searches that scale, and tests that find v-structures

PART I · STRUCTURE LEARNING

11.1 Why Structure Learning Is Hard

Proposition 11.1 (The DAG space explodes). *The number $a(n)$ of distinct DAGs on n labeled nodes grows super-exponentially in n (Robinson's recurrence, Section 11.20); already at $n = 8$ nodes there are $a(8) = 783{,}702{,}329{,}343 > 7.8 \times 10^{11}$ possible DAGs (Fig. 11.2, left), compared with $2^{\binom{8}{2}} = 2^{28} \approx 2.7 \times 10^8$ undirected graphs (`ex_dag_count()`: $a(1), \dots, a(5) = 1, 3, 25, 543, 29281$). More precisely, $2^{n(n-1)/2} \le a(n) \le n!\, 2^{n(n-1)/2}$, so $\log a(n)$ grows like n^2.*

Proof. Fix an ordering of the nodes. Each of the $\binom{n}{2} = n(n-1)/2$ pairs may or may not carry an edge pointing forward in that ordering, and every such choice is acyclic: this gives $2^{n(n-1)/2}$ distinct DAGs, the lower bound. Every DAG has at least one topological ordering, so every DAG is counted among the $n!$ orderings times $2^{n(n-1)/2}$ edge choices, the upper bound. For $n = 8$: $2^{28} = 268{,}435{,}456 \le a(8) \le 8!\, 2^{28} \approx 1.08 \times 10^{13}$. □

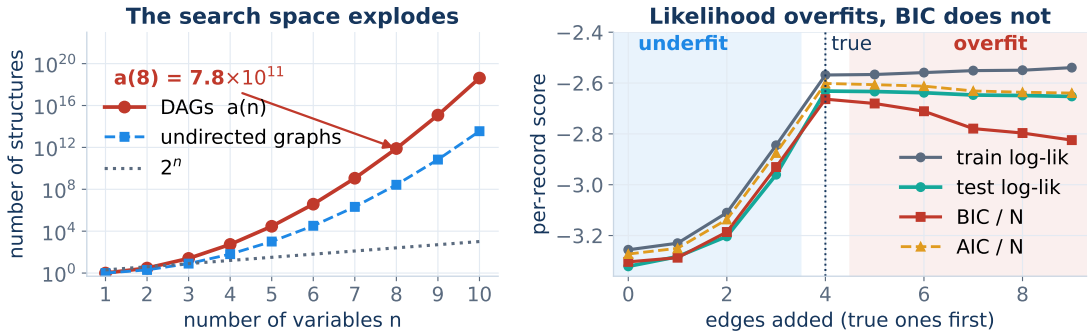


Figure 11.2. Left: the number of candidate DAG structures explodes far faster than the number of variables. Right: per-record scores along the nested DAGs of the score-curve paragraph in Section 11.2 (student network of Worked Example 11.3; $N = 300$ training and 20,000 test records; the four true edges are added first). Training log-likelihood never decreases, while test log-likelihood, BIC$/N$ and AIC$/N$ all peak at the true 4-edge graph (`ex_score_curve()`).

Two broad families of practical structure-learning methods exist:

- **Score-based search**: define a score (e.g., BIC, which penalizes the log-likelihood by $\frac{\log N}{2}$ times the number of free parameters, N being the number of records; defined in (11.2) directly generalizing the MLE machinery of Session 10) and search the DAG space greedily (add/remove/reverse one edge at a time) rather than exhaustively.
- **Constraint-based search**: run a sequence of conditional-independence tests (the sample analogue of Session 5's definition of conditional independence) to infer which edges *must* be absent, then orient the remaining edges using the collider-detection logic from Session 6 (a v-structure $X \to Z \leftarrow Y$ is distinguishable from a chain precisely because X and Y test as marginally independent but as dependent given Z, whereas in a chain or fork the pattern is reversed).

Finding the highest-scoring DAG is NP-hard even when every node may have at most two parents (Chickering 1996), so both families rely on heuristics or on assumptions. In practice, on real applied problems, structure is very often *not* learned purely from data at all: an expert-elicited graph (like our student network or a clinical DAG built from established medical knowledge) is preferred whenever available, both because data-driven structure search is statistically unreliable at

modest sample sizes and — as Session 12 will make sharply clear — because a graph learned purely from *observational, correlational* structure can be silent or actively misleading about which edges represent genuine causal mechanisms versus mere statistical association. We will not resolve that tension today; it is the very question the second half of these notes (Sessions 12–15) is devoted to.

PART I · STRUCTURE LEARNING

11.2 Scoring Structures: Model Selection

Fix a DAG $\mathcal{G}$ over n discrete variables, with r_i values for X_i and q_i configurations of its parents $U_i = \mathrm{pa}(X_i)$. With complete data $\mathcal{D}$ of N records, let $N_{x,u}$ be the number of records with $X_i = x$ and $U_i = u$ and $N_u = \sum_x N_{x,u}$; Session 10 gave the MLE $\hat\theta_{x|u} = N_{x,u}/N_u$. Throughout, log and ln both denote the natural logarithm.

Proposition 11.2 (Likelihood = mutual information)**.** *The maximised log-likelihood decomposes over families as*

$$\ell(\hat\theta_{\mathcal{G}} : \mathcal{D}) = N \sum_i \hat I(X_i; U_i) - N \sum_i \hat H(X_i), \tag{11.1}$$

where $\hat I$ and $\hat H$ are mutual information and entropy of the empirical distribution $\hat P$. Consequently adding a parent never decreases the likelihood, and the complete DAG always maximises it.

Proof. $\ell = \sum_i \sum_{x,u} N_{x,u} \log \hat P(x \mid u) = -N \sum_i \hat H(X_i \mid U_i)$, and $\hat H(X_i \mid U_i) = \hat H(X_i) - \hat I(X_i; U_i)$. By the chain rule of mutual information, $\hat I(X; U \cup \{Z\}) = \hat I(X; U) + \hat I(X; Z \mid U) \geq \hat I(X; U)$, because conditional mutual information is a (conditional) KL divergence, hence non-negative (Session 5). □

Empirical mutual information is almost never 0, so likelihood adds every edge: it *overfits*. Practical scores add a counterweight (Table 11.1). Writing $d = \dim(\mathcal{G}) = \sum_i (r_i - 1) q_i$ for the number of free parameters (Session 5's parameter count; for binary variables a node with k parents contributes 2^k):

$$\mathrm{AIC}(\mathcal{G}) = \ell(\hat\theta_{\mathcal{G}}) - d, \qquad \mathrm{BIC}(\mathcal{G}) = \ell(\hat\theta_{\mathcal{G}}) - \tfrac{\log N}{2}\, d. \tag{11.2}$$

The *Bayesian score* is the log marginal likelihood $\log P(\mathcal{D} \mid \mathcal{G}) = \log \int P(\mathcal{D} \mid \theta, \mathcal{G})\, p(\theta \mid \mathcal{G})\, d\theta$. With independent Dirichlet priors $\mathrm{Dir}(\alpha_{1|u}, \ldots, \alpha_{r_i|u})$ on every CPT row (global and local parameter independence, Session 10) the integral factorises in closed form (Heckerman, Geiger & Chickering 1995):

$$P(\mathcal{D} \mid \mathcal{G}) = \prod_i \prod_u \frac{\Gamma(\alpha_u)}{\Gamma(\alpha_u + N_u)} \prod_x \frac{\Gamma(\alpha_{x|u} + N_{x,u})}{\Gamma(\alpha_{x|u})}, \qquad \alpha_u = \textstyle\sum_x \alpha_{x|u}. \tag{11.3}$$

All $\alpha_{x|u} = 1$ gives the *K2* score; $\alpha_{x|u} = \alpha/(r_i q_i)$ gives *BDeu* with equivalent sample size α. Unlike the likelihood, (11.3) penalises complexity automatically: a richer graph spreads its prior over more parameters (the Bayesian Occam's razor). Finally, *cross-validation* scores a structure by the log-likelihood of held-out data; per record this estimates $\mathbb{E}_{P^*} \log \hat P_{\mathcal{G}}(\mathbf{X}) = -H(P^*) - \mathrm{KL}(P^* \| \hat P_{\mathcal{G}})$, i.e. minus the predictive KL divergence up to the constant entropy $H(P^*)$ of the true distribution P^*.

The scores at work. Figure 11.2 (right) uses the binary student network of Worked Example 11.3 ($D \to G \leftarrow I$, $I \to S$, $G \to L$). From $N = 300$ training records we fit a nested sequence of DAGs that adds the true edges $D \to G$, $I \to G$, $I \to S$, $G \to L$ and then the spurious edges $D \to L$, $I \to L$, $S \to L$, $D \to S$, $S \to G$, and evaluate each on 20,000 fresh test records (test CPTs use add-0.5 smoothing to avoid $\log 0$). The parameter count rises $d = 5, 6, 8, 9, 10, 12, 16, 24, 26, 30$. The training log-likelihood per record rises monotonically from -3.2560 (empty) to $-3.2301, \ldots,$ -2.5683 (true graph) and on to -2.5393 (all nine edges), as Proposition 11.2 guarantees; the test log-likelihood peaks at the true graph (-2.6317, versus -2.6334 with one spurious edge and -2.6526 with all five), and so do BIC/N (-2.6633) and AIC/N (-2.6016).

PYTHON LAB · `s11_examples.py` · `ex_score_curve()`

```
edges=0 k= 5 train=-3.2560 test=-3.3212 | edges=4 k=10 train=-2.5683
test=-2.6317 BIC/N=-2.6633 AIC/N=-2.6016
edges=9 k=30 train=-2.5393 test=-2.6526   BIC picks 4, AIC picks 4, test-LL
best at 4
```

Table 11.1. Model-selection criteria for Bayesian-network structures ($d = \dim \mathcal{G}$).

Score	Complexity penalty	Behaviour as $N \to \infty$	Remarks
Log-likelihood	none	picks the complete DAG	(11.1): never use alone
AIC	d	keeps a spurious binary edge w.p. $\to P(\chi^2_1 > 2) = 0.157$	good for prediction, not identification
BIC / MDL	$\frac{d}{2} \log N$	consistent (Proposition 11.9)	Laplace approximation of the Bayesian score
BDe / BDeu, K2	implicit, via the prior	consistent; BDeu score-equivalent	closed form (11.3); prior strength α matters
Held-out log-lik	implicit	estimates predictive KL	costly; needs enough data to split

All five scores are *decomposable*: a sum of family scores $\sum_i \mathrm{FamScore}(X_i, U_i : \mathcal{D})$, each depending only on the counts of X_i and its parents. Likelihood, AIC, BIC and BDeu are also *score equivalent*: Markov-equivalent DAGs (Theorem 11.1) get the same score (they have the same maximised likelihood, Worked Example 11.19, and the same number of parameters; K2 is the exception, Worked Example 11.2).

Worked Example 11.1 (A BIC calculation, structure search made concrete (instructor supplement))**.** A structure-learning algorithm compares two candidate structures for (A, B) using $N = 100$ fully-observed records: 60 have A=1, 40 have A=0; among A=1, B=1 occurs 48 times; among A=0, B=1 occurs 8 times. Compare $\mathrm{BIC} = \ell(\hat\theta) - \frac{d}{2} \log N$ for the edge model ($A \to B$) versus the no-edge model ($A \perp B$).

Solution. **Edge model.** MLEs: $\hat P(A{=}1) = 0.6$, $\hat P(B{=}1 \mid A{=}1) = 48/60 = 0.8$, $\hat P(B{=}1 \mid A{=}0) = 8/40 = 0.2$. Log-likelihood: $\ell_{\text{edge}} = 60 \ln 0.6 + 40 \ln 0.4 + 48 \ln 0.8 + 12 \ln 0.2 + 8 \ln 0.2 + 32 \ln 0.8$. (the four B-counts are $N_{B=1,A=1} = 48$, $N_{B=0,A=1} = 60 - 48 = 12$, $N_{B=1,A=0} = 8$, $N_{B=0,A=0} = 40 - 8 = 32$). Grouping the B-terms ($48+32 = 80$ occurrences of $\ln 0.8$, $12+8 = 20$ of $\ln 0.2$):

$$\begin{aligned} \ell_{\text{edge}} &= 60(-0.5108) + 40(-0.9163) + 80(-0.2231) + 20(-1.6094) \\ &= -30.65 - 36.65 - 17.85 - 32.19 = -67.30 - 50.04 = \mathbf{-117.34}. \end{aligned}$$

With $d_{\text{edge}} = 3$ free parameters (one for $P(A)$, two for $P(B \mid A)$) and $\ln 100 = 4.6052$, $\text{BIC}_{\text{edge}} = -117.34 - 1.5 \cdot 4.6052 = -117.34 - 6.91 = \mathbf{-124.25}$.
No-edge model. $\hat{P}(B{=}1) = (48+8)/100 = 0.56$ (pooled). $\ell_{\text{indep}} = [60 \ln 0.6 + 40 \ln 0.4] + [56 \ln 0.56 + 44 \ln 0.44] = -67.30 + (-32.47 - 36.12) = -67.30 - 68.59 = \mathbf{-135.89}$. With $d_{\text{indep}} = 2$, $\text{BIC}_{\text{indep}} = -135.89 - \frac{2}{2} \cdot 4.6052 = \mathbf{-140.50}$.
Since $-124.25 > -140.50$ by a wide margin ($-124.25 - (-140.50) = 16.25$), the edge model wins decisively: the data's strong dependence (0.8 vs. 0.2) far outweighs the extra parameter's complexity penalty at $N = 100$. AIC agrees: $-117.34 - 3 = -120.34$ against $-135.89 - 2 = -137.89$. By (11.1) the likelihood gain $-117.34 - (-135.89) = 18.55$ is exactly $N\hat{I}(A;B)$. The reversed DAG $B \to A$ has the B-family $56 \ln 0.56 + 44 \ln 0.44 = -68.59$ and the $A \mid B$ family $48 \ln \frac{48}{56} + 8 \ln \frac{8}{56} + 12 \ln \frac{12}{44} + 32 \ln \frac{32}{44} = -48.75$, in total the same -117.34: data cannot orient a single edge. (An earlier edition of these notes gave -135.87 and -140.48; the values above are recomputed.)

Worked Example 11.2 (The Bayesian score of the same data)**.** Score the two structures of Worked Example 11.1 by the K2 marginal likelihood (11.3).
Solution. For one binary family with a uniform Beta$(1,1)$ prior, (11.3) is $\frac{n_1!\, n_0!}{(n_1+n_0+1)!}$. The A-factor $\ln \frac{60!\,40!}{101!} = -69.41$ is common. No edge: B contributes $\ln \frac{56!\,44!}{101!} = -70.68$; edge: $\ln \frac{48!\,12!}{61!} + \ln \frac{8!\,32!}{41!} = -32.08 - 21.87$ (log-factorials via $\ln n! = \texttt{lgamma}(n+1)$). Hence

$$\log P(\mathcal{D} \mid \text{no edge}) = -69.41 - 70.68 = -140.09,$$
$$\log P(\mathcal{D} \mid A \to B) = -69.41 - 32.08 - 21.87 = -123.36,$$

a log Bayes factor of $-123.36 - (-140.09) = 16.73$ (Bayes factor $e^{16.73} = 1.85 \times 10^7$) — remarkably close to the BIC difference 16.25 of Worked Example 11.1; the gap $16.73 - 16.25 = 0.48$ is the $O(1)$ term of the Laplace approximation (box "BIC as a Laplace approximation" in Section 11.15). K2 is not score equivalent. BDeu with $\alpha = 2$ puts Beta$(1,1)$ on a parentless binary node ($\alpha/(r_i q_i) = 2/2$) and Beta$(\frac{1}{2}, \frac{1}{2})$ on each row of a binary node with one binary parent ($2/4$); it gives -123.803 to *both* $A \to B$ and $B \to A$ (and -140.090 to the empty graph).

PYTHON LAB · `s11_examples.py` · `ex_bic_two_node()`, `ex_bde_score()`

```
BIC edge = -124.25, indep = -140.50 (diff 16.25); AIC -120.34 vs -137.89;
N*I_hat = 18.55; B->A loglik -117.34
K2 factors:  -69.41, -70.68, -32.08, -21.87; log P(D|no edge) = -140.09, log
P(D|A->B) = -123.36
log BF = 16.73 (BF = 1.853e+07); BDeu(alpha=2):  no edge -140.090, A->B =
B->A = -123.803
```

PART I · STRUCTURE LEARNING

11.3 Score-Based Search: Greedy Hill Climbing

Decomposability is what makes search affordable: a local move changes the parent set of one node (add or delete $u \to v$) or two nodes (reverse $u \to v$), so its score change Δ needs only one or two family scores, which can be cached.

Greedy hill climbing (1) starts from the empty graph (or a random DAG, for a restart) and caches family scores; (2) evaluates every legal move (add, delete or reverse one edge, keeping acyclicity and an in-degree bound) through the one or two changed families; (3) applies the best move whose result is not on a short *tabu list* of recently visited graphs; (4) stops when no move improves the score, and (5) repeats from several random starts, returning the best local optimum. Each step examines $O(n^2)$ moves; local optima and plateaus of Markov-equivalent graphs motivate tabu lists and restarts, and searching over equivalence classes (GES) is a refinement (Koller & Friedman 2009, Ch. 18).

Worked Example 11.3 (Hill climbing recovers the student network)**.** The *student network* of Session 6, in a hypothetical all-binary version: Difficulty D (1 = hard), Intelligence I (1 = high), Grade G (1 = good), SAT S (1 = high), Letter L (1 = strong), with DAG $D \to G \leftarrow I$, $I \to S$, $G \to L$ and illustrative CPTs $P(D{=}1) = 0.4$, $P(I{=}1) = 0.3$; $P(G{=}1 \mid d, i) = 0.5, 0.9, 0.2, 0.6$ for $(d, i) = (0,0), (0,1), (1,0), (1,1)$; $P(S{=}1 \mid i) = 0.1, 0.8$ for $i = 0, 1$; $P(L{=}1 \mid g) = 0.1, 0.85$ for $g = 0, 1$. Sample $N = 5000$ records (simulation, seed 2026) and hill-climb with BIC (at most 3 parents, tabu list of the last 10 graphs) from five starts: the empty graph and four random DAGs (random node order, each forward pair joined with probability 0.3).
Solution (simulation output). From the empty graph four moves reach BIC -13003.29, which is exactly the BIC of the true DAG, with exactly its edges; restart 4 also reaches it, while restarts 1–3 stop at worse local optima (-13029.95, -13014.62, -13007.31). The v-structure is oriented because reversing one of its arrows changes the equivalence class and the score (reversing $D \to G$ gives -13031.58); the direction of $I \to S$, in contrast, is an arbitrary tie-break, since $S \to I$ is Markov equivalent and has the same BIC -13003.29.

PYTHON LAB · s11_examples.py · ex_hill_climb()

```
restart 0:  BIC = -13003.29 after 4 moves, edges (D,G),(I,G),(I,S),(G,L) =
true DAG
restarts 1-4:  -13029.95, -13014.62, -13007.31, -13003.29; S->I instead of
I->S: -13003.29; G->D: -13031.58
```

PART I · STRUCTURE LEARNING

11.4 Constraint-Based Learning: The PC Algorithm

Definition 11.1 (Markov equivalence, CPDAG)**.** Two DAGs are *Markov equivalent* if they encode the same set of conditional independences. The *CPDAG* (essential graph) of an equivalence class has a directed edge where every member agrees on the orientation and an undirected edge elsewhere.

Theorem 11.1 (Verma & Pearl 1990)**.** *Two DAGs are Markov equivalent if and only if they have the same skeleton and the same v-structures (colliders $X \to Z \leftarrow Y$ with X, Y non-adjacent).*

Because observational data only reveal independences, *no* method — score-based or constraint-based — can do better than the CPDAG from observational data alone. The PC algorithm (Spirtes, Glymour & Scheines 2000) finds it with CI tests:

PC algorithm. 1 Start from the complete undirected graph. 2 For $\ell = 0, 1, 2, \ldots$ (while some node has more than ℓ neighbours): for each adjacent pair $X - Y$ and each $S \subseteq \mathrm{adj}(X) \setminus \{Y\}$ with $|S| = \ell$, if the test accepts $X \perp\!\!\!\perp Y \mid S$, delete $X - Y$ and store $\mathrm{sep}(X, Y) = S$. 3 For every non-adjacent pair X, Y with common neighbour $Z \notin \mathrm{sep}(X, Y)$, orient $X \to Z \leftarrow Y$. 4 Apply Meek's rules, e.g. R1: $X \to Z - W$ with X, W non-adjacent $\Rightarrow Z \to W$ (otherwise a new v-structure would appear).

Under *faithfulness* (every independence in P is a d-separation of the true DAG) and a perfect CI oracle, PC returns exactly the true CPDAG, in polynomial time for bounded degree. In practice the test is the likelihood-ratio G^2-test (discrete data, Session 5), with statistic $G^2 = 2N\hat{I}(X; Y \mid S)$, approximately χ^2_{df} under independence with $\mathrm{df} = (r_X - 1)(r_Y - 1)\prod_{Z \in S} r_Z$ (for binary variables $\mathrm{df} = 2^{|S|}$); or, for Gaussian data, Fisher's z-test (Session 5): with $\hat{r}$ the sample partial correlation of X, Y given S from n observations, $z = \frac{1}{2} \ln \frac{1+\hat{r}}{1-\hat{r}} \cdot \sqrt{n - |S| - 3}$ is compared with $N(0, 1)$ quantiles.

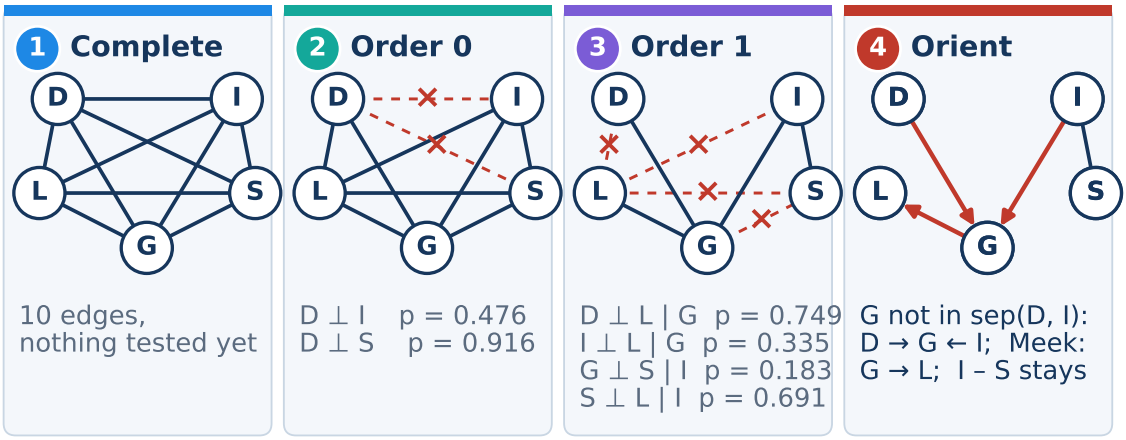


Figure 11.3. The PC algorithm on $N = 5000$ records from the student network of Worked Example 11.3. Red dashed edges are deleted at each order, with the G^2-test p-values; the final panel is the CPDAG: $D \to G \leftarrow I$ and $G \to L$ are compelled, $I - S$ cannot be oriented (`ex_pc_algorithm()`).

Worked Example 11.4 (PC on the student data)**.** Run PC with G^2-tests at level 0.01 on a fresh sample of $N = 5000$ records from the network of Worked Example 11.3 (simulation, seed 2027). A pair is declared independent when $p > 0.01$.

Solution (Fig. 11.3; simulation output). 1 Order 0 removes $D - I$ ($p = 0.476$) and $D - S$ ($p = 0.916$); every other marginal test rejects (e.g. $G^2 = 257.59$, $\mathrm{df} = 1$, $p = 5.8 \times 10^{-58}$ for D–L). 2 Order 1 removes $D - L$ ($p = 0.749$) and $I - L$ ($p = 0.335$) given G, and $G - S$ ($p = 0.183$) and $S - L$ ($p = 0.691$) given I. 3 D and I are non-adjacent with common neighbour $G \notin \mathrm{sep}(D, I) = \emptyset$, so $D \to G \leftarrow I$ — the collider signature $D \perp\!\!\!\perp I$ but $D \not\perp\!\!\!\perp I \mid G$ ($G^2 = 65.87$, $\mathrm{df} = 2$, $p = 5.0 \times 10^{-15}$). 4 Meek R1 with $D \to G - L$ and D, L non-adjacent gives $G \to L$; $I - S$ stays undirected, since no arrow points into I or S and $I \to S$, $S \to I$ are Markov equivalent.

PYTHON LAB · `s11_examples.py` · `ex_pc_algorithm()`

```
D _||_ I: G = 0.51, p = 0.476; D _||_ I | G: G = 65.87, df = 2, p = 4.98e-15;
D _||_ L: G = 257.59
removed D-I, D-S (order 0); D-L, I-L | G and G-S, S-L | I (order 1; p =
0.749, 0.335, 0.183, 0.691)
CPDAG directed:  (D,G),(G,L),(I,G); undirected:  (I,S)
```

Worked Example 11.5 (Structure learning sanity check)**.** Given complete data on the student network's five variables, a constraint-based structure learner tests marginal and conditional independences between every pair. It should find $D \perp\!\!\!\perp I$ (no edge and no common cause), and — critically, per Session 6's analysis of colliders — it should find D and I marginally independent *but* dependent given G. A correct constraint-based algorithm uses exactly this asymmetric pattern to infer that G must be a collider with parents D and I (rather than, say, a chain), recovering the correct edge orientations from data alone (Worked Example 11.4 confirmed this numerically: $p = 0.476$ marginally, $p = 4.98 \times 10^{-15}$ given G) — but notice that this recovers only the graph's *independence skeleton and colliders*; whether an edge like $D \to G$ represents difficulty genuinely causing the grade, versus a purely associational convention, is a question this algorithm has no way to answer. That distinction is exactly what Session 12 forces us to confront.

PART II

Sampling-Based Inference

Replace sums by averages over random draws — and pay attention to the weights

PART II · MONTE CARLO

11.5 When Exact Inference Fails: The Case for Sampling

Recall Session 7's headline result: for n variables with at most r values each, VE costs $O(n \cdot r^{w+1})$ where w is the treewidth. Many real graphical models — densely connected social networks, image grids with strong neighbor coupling, gene regulatory networks — have treewidth comparable to n itself, making exact inference computationally hopeless regardless of how cleverly we order eliminations. *Sampling-based approximate inference* trades exactness for tractability: instead of computing $P(y \mid e)$ exactly, we draw samples from (a distribution close to) P, and estimate $P(y \mid e)$ by counting.

The Monte Carlo principle. Any query is an expectation: $P(y) = \mathbb{E}_P[\mathbb{1}\{Y = y\}]$. Given $\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)} \overset{\text{iid}}{\sim} P$,

$$\hat{\mu}_N = \frac{1}{N} \sum_{m=1}^{N} f(\mathbf{x}^{(m)}) \;\approx\; \mu = \mathbb{E}_P[f(\mathbf{X})], \qquad \mathrm{SE}(\hat{\mu}_N) = \frac{\sigma_f}{\sqrt{N}}, \quad \sigma_f^2 = \mathrm{Var}_P f(\mathbf{X}), \tag{11.4}$$

whatever the dimension of $\mathbf{x}$ (Theorem 11.2). For a probability query $f = \mathbb{1}\{Y = y\}$, $\sigma_f^2 = p(1-p)$ with $p = P(y)$. The error decays like $1/\sqrt{N}$ (Fig. 11.4, left): each extra digit of accuracy costs $100\times$ more samples.

11.5.1 Forward (Ancestral) Sampling

Definition 11.2 (Ancestral sampling)**.** Given a Bayesian network with no evidence, sample every variable in an order consistent with the DAG (parents before children, always possible since the graph is acyclic): draw $x_i \sim P(X_i \mid \mathrm{pa}(X_i))$ with the parents set to their already-sampled values.

This produces exact i.i.d. samples from the joint $P(\mathbf{x})$ with no approximation at all — it is simply the BN chain rule $P(\mathbf{x}) = \prod_i P(x_i \mid \mathrm{pa}_i)$ (Session 6) turned into a generative procedure. The catch appears once we need to condition on evidence: if evidence $E = e$ is rare, most sampled trajectories will simply not match e and must be discarded, wasting nearly all the computation (*rejection sampling*). *Importance sampling* improves on this by sampling from a more convenient proposal distribution Q and reweighting each sample by $P(\mathbf{x})/Q(\mathbf{x})$ rather than discarding mismatches outright — a technique we will meet again, doing essentially the same reweighting job, as inverse propensity weighting in Session 15.

Worked Example 11.6 (Monte Carlo on the sprinkler network)**.** The sprinkler network of Russell & Norvig (2021): $C \to S$, $C \to R$, $(S, R) \to W$ with $P(c) = 0.5$, $P(s \mid c) = 0.1$, $P(s \mid \neg c) = 0.5$, $P(r \mid c) = 0.8$, $P(r \mid \neg c) = 0.2$, $P(w \mid s, r) = 0.99$, $P(w \mid s, \neg r) = P(w \mid \neg s, r) = 0.90$, $P(w \mid \neg s, \neg r) = 0$. Estimate $P(W{=}1)$ from $N = 1000$ ancestral samples.
Solution. *Exact value by enumeration.* First $P(s, r) = \sum_c P(c)P(s \mid c)P(r \mid c)$: $P(S{=}1, R{=}1) = 0.5(0.1 \cdot 0.8) + 0.5(0.5 \cdot 0.2) = 0.04 + 0.05 = 0.09$, $P(S{=}1, R{=}0) = 0.5(0.1 \cdot 0.2) + 0.5(0.5 \cdot 0.8) = 0.01 + 0.20 = 0.21$, $P(S{=}0, R{=}1) = 0.5(0.9 \cdot 0.8) + 0.5(0.5 \cdot 0.2) = 0.36 + 0.05 = 0.41$, $P(S{=}0, R{=}0) = 0.5(0.9 \cdot 0.2) + 0.5(0.5 \cdot 0.8) = 0.09 + 0.20 = 0.29$. Then

$$\begin{aligned} P(W{=}1) &= \textstyle\sum_{s,r} P(s, r)P(W{=}1 \mid s, r) = 0.09 \cdot 0.99 + 0.21 \cdot 0.90 + 0.41 \cdot 0.90 + 0.29 \cdot 0 \\ &= 0.0891 + 0.1890 + 0.3690 = 0.6471. \end{aligned}$$

Monte Carlo (one simulated run, seed 2026). The sample proportion is $\hat{p} = 0.6330$ with $\mathrm{SE} = \sqrt{\hat{p}(1-\hat{p})/N} = \sqrt{0.6330 \cdot 0.3670/1000} = 0.0152$, so the 95% interval $0.6330 \pm 1.96 \cdot 0.0152 = [0.6031, 0.6629]$ covers the truth. *Planning.* Since $\sigma^2 = 0.6471 \cdot 0.3529 = 0.2284$, $\sigma = 0.478$, the SE $\sigma/\sqrt{N}$ is 0.048, 0.0048, 0.00048 at $N = 10^2, 10^4, 10^6$; an SE of 0.005 needs $N = \sigma^2/0.005^2 = 0.2284/0.000025 = 9134.5$, i.e. $N = 9135$ samples.

PYTHON LAB · `s11_examples.py` · `ex_mc_basics()`

```
exact P(W=1) = 0.6471; N = 1000:  estimate = 0.6330, SE = 0.0152, 95% CI =
[0.6031, 0.6629]; N for SE <= 0.005:  9,135
```

PART II · MONTE CARLO

11.6 Rejection Sampling and Likelihood Weighting

Rejection sampling for $P(y \mid e)$ draws ancestral samples and keeps those with $E = e$; the kept samples are exact draws from $P(\cdot \mid e)$, but the acceptance rate is $P(e)$, so the expected number of draws per kept sample is $1/P(e)$, and $P(e)$ typically shrinks exponentially in the number of evidence variables (Worked Examples 11.17 and 11.18).

Likelihood weighting (LW) never rejects: it clamps the evidence variables, samples the others

ancestrally, and gives each sample the weight

$$w(\mathbf{x}) = \prod_{i:\, X_i \in E} P(e_i \mid \mathrm{pa}(X_i)), \qquad \widehat{P}(y \mid e) = \frac{\sum_m w^{(m)} \mathbb{1}\{y^{(m)} = y\}}{\sum_m w^{(m)}}. \tag{11.5}$$

LW is importance sampling (next section) with the proposal "sample non-evidence variables from their CPTs": the ratio $P(\mathbf{x}, e)/Q(\mathbf{x})$ leaves exactly the evidence CPTs. Its weakness: evidence only influences its *descendants*' sampling; upstream variables are sampled from the prior and merely re-weighted.

Worked Example 11.7 (Rejection versus likelihood weighting)**.** In the sprinkler network, estimate $P(R{=}1 \mid S{=}1, W{=}1)$ with $N = 10{,}000$ samples each way.
Solution. Exact value, with the $P(s, r)$ of Worked Example 11.6: $P(S{=}1, R{=}1, W{=}1) = 0.09 \cdot 0.99 = 0.0891$ and $P(S{=}1, R{=}0, W{=}1) = 0.21 \cdot 0.90 = 0.1890$, so $P(S{=}1, W{=}1) = 0.0891 + 0.1890 = 0.2781$ and $P(R{=}1 \mid S{=}1, W{=}1) = 0.0891/0.2781 = 0.3204$. The sprinkler explains away the wet grass: without $S{=}1$, $P(R{=}1, W{=}1) = 0.0891 + 0.41 \cdot 0.90 = 0.4581$ and $P(R{=}1 \mid W{=}1) = 0.4581/0.6471 = 0.7079$. *Simulation* (seed 2026). Rejection keeps 2852 of 10,000 samples (28.5%, close to $P(S{=}1, W{=}1) = 27.8\%$), estimate 0.3310. LW samples C and R from their CPTs and weights each sample by $P(S{=}1 \mid c)\, P(W{=}1 \mid S{=}1, r)$: $0.1 \cdot 0.99 = 0.099$, $0.1 \cdot 0.90 = 0.090$, $0.5 \cdot 0.99 = 0.495$, $0.5 \cdot 0.90 = 0.450$ for $(C, R) = (1,1), (1,0), (0,1), (0,0)$. All samples are used, the estimate is 0.3301, and the effective sample size $\mathrm{ESS} = (\sum w)^2 / \sum w^2$ ((11.7), next section) is 7012.

PYTHON LAB · `s11_examples.py` · `ex_rejection_lw()`

```
exact P(R=1 | S=1, W=1) = 0.3204; rejection:  2852 of 10000 kept (28.5%),
estimate 0.3310
likelihood weighting:  estimate = 0.3301; ESS = 7012 (70.1%)
```

PART II · MONTE CARLO

11.7 Importance Sampling and the Effective Sample Size

For any proposal q with $q(x) > 0$ wherever $f(x)p(x) \neq 0$,

$$\mathbb{E}_p[f(X)] = \mathbb{E}_q\big[w(X) f(X)\big], \quad w = \frac{p}{q}; \qquad \hat{\mu}_{\mathrm{IS}} = \frac{1}{N}\sum_m w^{(m)} f^{(m)}, \quad \hat{\mu}_{\mathrm{SNIS}} = \frac{\sum_m w^{(m)} f^{(m)}}{\sum_m w^{(m)}}. \tag{11.6}$$

The self-normalised form only needs p up to a constant (posteriors, Markov networks) and is consistent but slightly biased. The variance of $\hat{\mu}_{\mathrm{IS}}$ is governed by $\mathbb{E}_q[w^2 f^2]$ (Theorem 11.3); a heavy-tailed weight distribution means a few samples dominate. The standard diagnostic is

$$\mathrm{ESS} = \frac{\big(\sum_m w^{(m)}\big)^2}{\sum_m \big(w^{(m)}\big)^2} \in [1, N], \tag{11.7}$$

the number of equally weighted samples with the same variance (for f that is roughly constant); for $N \to \infty$, $\mathrm{ESS}/N \to 1/\mathbb{E}_q[w^2]$ when $\mathbb{E}_q w = 1$.

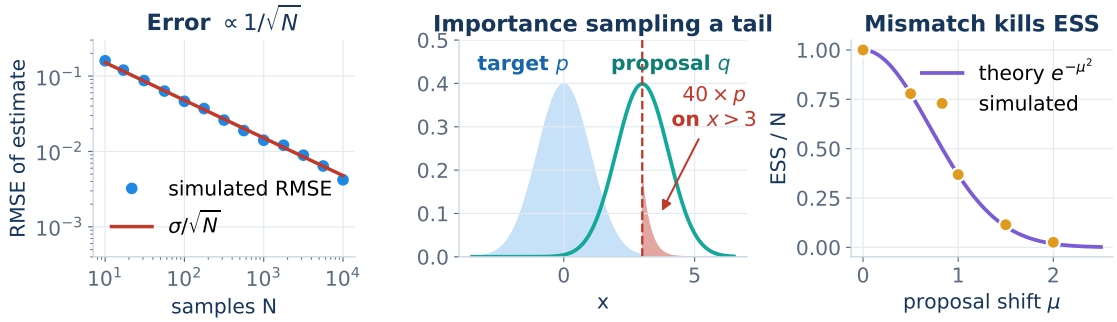


Figure 11.4. Left: the RMSE of $\hat{P}(W{=}1)$ in the sprinkler network of Worked Example 11.6 follows $\sigma/\sqrt{N}$ with $\sigma = 0.478$ (300 simulated replications per N). Middle: to estimate $P(X > 3)$ under $N(0,1)$ we sample from $N(3,1)$, where the event is common; the red sliver is the target density on $x > 3$, magnified 40 times to be visible. Right: for target $N(0,1)$ and proposal $N(\mu,1)$, $\mathrm{ESS}/N \to e^{-\mu^2}$ (Proposition 11.5); dots are simulations with $N = 200{,}000$.

Worked Example 11.8 (A tail probability)**.** Estimate $p = P(X > 3)$, $X \sim N(0,1)$, with $N = 10{,}000$ draws.

Solution ($\phi, \bar{\Phi}$: standard normal density and upper tail $\bar{\Phi}(c) = P(X > c)$; simulations with seed 2026). $p = \bar{\Phi}(3) = 1.3499 \times 10^{-3}$. Naive MC sees only 16 hits (estimate $16/10{,}000 = 1.6 \times 10^{-3}$); its SE $\sqrt{p(1-p)/N} = 3.672 \times 10^{-4}$ is 27.2% of p. With $q = N(3,1)$, $w(y) = \phi(y)/\phi(y-3) = \exp\{-\frac{1}{2}y^2 + \frac{1}{2}(y-3)^2\} = e^{-3y+4.5}$, and the IS estimate is 1.3394×10^{-3}. By Proposition 11.5 with $\mu = c = 3$, the exact per-sample variance of wf is

$$e^9 \bar{\Phi}(6) - p^2 = 7.9944 \times 10^{-6} - 1.8222 \times 10^{-6} = 6.1722 \times 10^{-6},$$

against $p(1-p) = 1.3481 \times 10^{-3}$ for naive MC — a $1.3481 \times 10^{-3}/6.1722 \times 10^{-6} = 218$-fold reduction (IS relative SE $\sqrt{6.1722 \times 10^{-6}/10^4}/p = 1.84\%$). Minimising $e^{\mu^2}\bar{\Phi}(3+\mu) - p^2$ over a grid of shifts gives $\mu^* = 3.16$, close to 3. The zero-variance proposal is the truncated normal $\phi(x)\mathbb{1}\{x > 3\}/p$ (Theorem 11.3), which needs p itself. The self-normalised estimate, however, is 1.738×10^{-3} (ESS 44): $N(3,1)$ covers the bulk of the target badly (the 10,000 weights average 0.77 instead of their expectation 1), so when p is normalised use $\hat{\mu}_{\mathrm{IS}}$. Even ESS 44 is optimistic: Proposition 11.5 gives ESS $\approx Ne^{-9} = 1.23$, because $\mathbb{E}_q[w^2] = e^9$ is dominated by draws far in the left tail of q that a sample of 10,000 practically never contains.

PYTHON LAB · s11_examples.py · ex_importance_tail(), ex_ess_shift()

```
naive:  1.6000e-03 (rel.  SE 27.2%); IS: 1.3394e-03 (rel.  SE 1.84%); e^9
Phibar(6) = 7.9944e-06; p^2 = 1.8222e-06
factor 218; mu* = 3.16; SNIS 1.7380e-03, ESS 44; mean weight 0.7707; N e^-9
= 1.23
ESS/N for mu = 0.5, 1.0, 1.5:  0.778, 0.369, 0.115   theory exp(-mu^2):
0.779, 0.368, 0.105
```

Worked Example 11.9 (Self-normalised IS and the ESS by hand)**.** Five draws from a proposal have importance weights $w = (0.5, 2.5, 1.0, 0.25, 0.75)$ and function values $f = (3, 1, 2, 6, 2)$. Compute $\hat{\mu}_{\mathrm{IS}}$, $\hat{\mu}_{\mathrm{SNIS}}$ and the ESS.

Solution. $\sum w = 0.5 + 2.5 + 1.0 + 0.25 + 0.75 = 5.00$; $\sum wf = 1.5 + 2.5 + 2.0 + 1.5 + 1.5 = 9.00$; $\sum w^2 = 0.25 + 6.25 + 1 + 0.0625 + 0.5625 = 8.125$. Hence $\hat{\mu}_{\mathrm{IS}} = 9.00/5 = 1.800$ and $\hat{\mu}_{\mathrm{SNIS}} = 9.00/5.00 = 1.800$ (they coincide only because these weights happen to average exactly 1), and $\mathrm{ESS} = 5.00^2/8.125 = 3.077$ of $n = 5$. The normalised weights $(0.10, 0.50, 0.20, 0.05, 0.15)$

show why: half of the total weight sits on one draw.

PYTHON LAB · s11_examples.py · ex_ess_hand()

```
sum w = 5.00; sum w f = 9.00; sum w^2 = 8.1250; IS = 1.800; SNIS = 1.8000;
ESS = 3.077 of n = 5
```

PART III

MCMC, Variational Inference and Particle Filters

Walk instead of jump; optimise instead of sample; filter instead of smooth

PART III · MCMC AND BEYOND

11.8 Markov Chains and Gibbs Sampling

When even constructing a good proposal Q is hard (common for densely-connected Markov networks, where there is no natural "ancestral" order at all), we turn to Markov chain Monte Carlo (MCMC): build a Markov chain over the state space whose *stationary distribution* is exactly the target P, then run it long enough and treat the states visited as (correlated, but eventually representative) samples from P.

Definition 11.3 (Stationarity and detailed balance)**.** A transition kernel $T(\mathbf{x} \to \mathbf{x}')$ has *stationary distribution* π if $\sum_{\mathbf{x}} \pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}')$ for all $\mathbf{x}'$. It satisfies *detailed balance* (is *reversible*) with respect to π if $\pi(\mathbf{x}) T(\mathbf{x} \to \mathbf{x}') = \pi(\mathbf{x}') T(\mathbf{x}' \to \mathbf{x})$ for all pairs.

Detailed balance implies stationarity (Proposition 11.6); for a finite irreducible, aperiodic chain the stationary distribution is unique and the chain converges to it from any start (box "Ergodic theorem and the Markov-chain CLT" in Section 11.14).

Algorithm 11.1 Gibbs Sampling

1: Initialize $\mathbf{x}^{(0)}$ arbitrarily.
2: **for** $t = 0, 1, 2, \ldots$ **do**
3: **for** $i = 1, \ldots, n$ (a fixed order; or pick i at random) **do**
4: Sample $x_i^{(t+1)} \sim P\big(X_i \mid x_1^{(t+1)}, \ldots, x_{i-1}^{(t+1)}, x_{i+1}^{(t)}, \ldots, x_n^{(t)}\big)$, the full conditional given the *current* values of all other variables (those already updated in this sweep at their new values).
5: **end for**
6: **end for**

The full conditional $P(X_i \mid x_{-i})$ has a beautiful simplification in a Markov network (Session 8): it only depends on X_i's *neighbors* in the graph, since every potential not involving X_i cancels in the normalisation — the neighbors form X_i's Markov blanket. This is another direct payoff of the undirected graphical structure; in a Bayesian network the blanket is parents, children and co-parents (Session 6), and $P(x_i \mid x_{-i}) \propto P(x_i \mid \mathrm{pa}_i) \prod_{j \in \mathrm{ch}(i)} P(x_j \mid \mathrm{pa}_j)$.

Proposition 11.3 (Gibbs sampling is correct, informally)**.** *The Markov chain defined by Gibbs sampling has the target distribution P as a stationary distribution: if $\mathbf{x}^{(t)} \sim P$, then $\mathbf{x}^{(t+1)} \sim P$ as well. If $P(\mathbf{x}) > 0$ for every state, it is the unique stationary distribution.*

The proof (Section 11.20) is a direct check of *detailed balance* for one coordinate update: in stationarity the chain is exactly as likely to move from state a to state b as from b to a, which is a sufficient condition for P to be stationary for that update; a sweep is a composition of such updates, each of which leaves P unchanged. Uniqueness needs every state to be reachable, which holds when all potentials are strictly positive. In practice, the chain needs some number of iterations ("burn-in") before it forgets its (usually poor) initialization and starts producing draws representative of P; Fig. 11.5 shows this concretely on the worked example below.

INTUITION

Do not confuse the number of Gibbs sweeps with the number of independent samples: consecutive draws are correlated (each differs from the last by only one variable's resampling). Practitioners discard a burn-in period and measure the correlation through the effective sample size (Definition 11.4). Thinning the chain (keeping only every k-th sample) makes the kept draws closer to independent, but it throws information away: it saves memory and never increases the ESS compared with keeping all draws (Section 11.10).

Worked Example 11.10 (Gibbs sampling a small Markov network, exactly checked)**.** Consider three binary spins $X_1, X_2, X_3 \in \{-1, +1\}$ arranged in a triangle (every pair connected), with pairwise potentials $\phi(x_i, x_j) = \exp(J\, x_i x_j)$, $J = 0.9$, favoring agreement (the same style of model as Session 8's grid, small enough here that we can also compute the exact answer by brute-force enumeration over all $2^3 = 8$ states, for comparison). *Exact distribution.* $P(\mathbf{x}) = e^{J(x_1x_2 + x_2x_3 + x_1x_3)}/Z$. In an all-agreeing state the three products are $+1$, giving weight $e^{3J} = e^{2.7} = 14.880$; in each of the six mixed states one spin disagrees with the other two, so the products sum to -1 and the weight is $e^{-J} = e^{-0.9} = 0.4066$. Hence $Z = 2 \cdot 14.880 + 6 \cdot 0.4066 = 32.199$, $P(+,+,+) = P(-,-,-) = 14.880/32.199 = 0.4621$ and each mixed state has $0.4066/32.199 = 0.0126$. *Simulation* (Fig. 11.5; Algorithm 11.1 with systematic scan, start $(+1, -1, +1)$, seed 7). We ran 3000 sweeps and discarded the first 50 as burn-in. The empirical frequencies of the remaining 2950 sweeps (0.4922 for $(+,+,+)$ and 0.4261 for $(-,-,-)$, every mixed state within 0.003 of 0.0126) match the exact values to within 0.0360. The residual error is Monte Carlo error inflated by autocorrelation: the chain switches between the two dominant states only rarely. This confirms Proposition 11.3 numerically without ever computing the partition function Z directly — exactly the situation we will face in larger, intractable models where brute-force checking is no longer available.

Worked Example 11.11 (A Gibbs full conditional, computed exactly (instructor supplement))**.** For the triangle spin model ($\phi(x_i, x_j) = e^{J x_i x_j}$, J=0.9, spins in $\{-1, +1\}$), compute the full conditional $P(X_1 \mid X_2{=}{+1}, X_3{=}{+1})$ used inside every sweep of Gibbs sampling.
Solution. X_1's only neighbors are X_2, X_3 (its Markov blanket in this graph), so $P(X_1 \mid X_2, X_3) \propto \phi(X_1, X_2)\phi(X_1, X_3)$. For $X_1{=}{+1}$: $e^{0.9(1)(1)}e^{0.9(1)(1)} = e^{1.8} \approx 6.0496$. For $X_1{=}{-1}$: $e^{0.9(-1)(1)}e^{0.9(-1)(1)} = e^{-1.8} \approx 0.1653$. Normalizing over the total $6.0496 + 0.1653 = 6.2149$:

$$P(X_1{=}{+1} \mid X_2{=}1, X_3{=}1) = \frac{6.0496}{6.2149} \approx \mathbf{0.9734}, \qquad P(X_1{=}{-1} \mid \cdots) \approx \mathbf{0.0266}.$$

This confirms, with real numbers, the claim that the full conditional depends only on X_1's

neighbors and requires normalizing over only the 2 values of X_1 itself — never the full $2^3 = 8$-state joint — which is exactly what makes Gibbs sampling cheap per step even when the joint distribution (and its partition function Z) is intractable to compute directly. In code the update is $P(X_i{=}{+1} \mid \cdot) = e^{h_i}/(e^{h_i} + e^{-h_i}) = 1/(1 + e^{-2h_i})$ with local field $h_i = J \sum_{j \neq i} x_j$ (here $h_1 = 1.8$ and $1/(1 + e^{-3.6}) = 0.9734$).

PYTHON LAB · s11_examples.py · ex_gibbs_triangle()

```
e^2.7 = 14.880; e^-0.9 = 0.4066; Z = 32.199; P(+++) = P(---) = 0.4621; mixed
= 0.0126; P(X1=+1 | +,+) = 0.9734
Gibbs (2950 kept sweeps):  +++ 0.4922, --- 0.4261; max |exact - Gibbs| =
0.0360
```

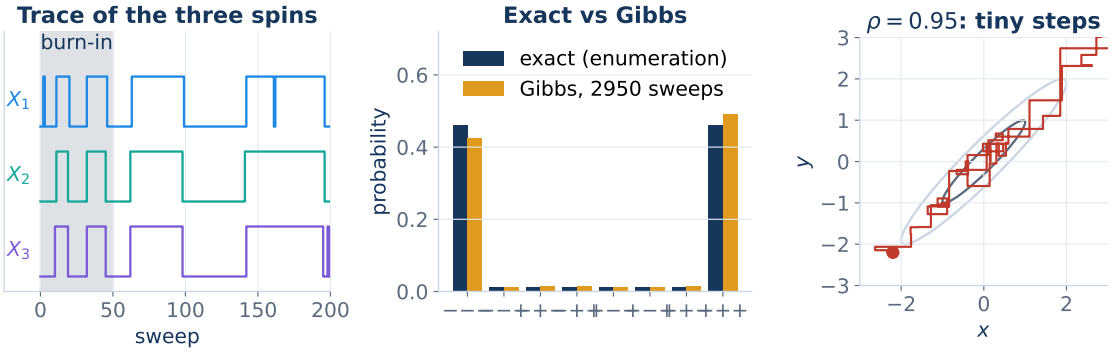


Figure 11.5. Left: trace of each spin variable across the first 200 Gibbs sweeps; the shaded region is discarded as burn-in. Middle: exact probabilities (brute-force enumeration) versus Gibbs sampler empirical frequencies, post burn-in. Right: Gibbs moves are axis-parallel, so on a bivariate normal with $\rho = 0.95$ it creeps along the ridge (40 sweeps from $(-2.2, -2.2)$).

PART III · MCMC AND BEYOND

11.9 Metropolis–Hastings

Gibbs needs full conditionals we can sample. Metropolis–Hastings (MH; Metropolis et al. 1953, Hastings 1970) needs only the ability to evaluate $\tilde{\pi} \propto \pi$ and to sample some proposal $q(\mathbf{x}' \mid \mathbf{x})$.

Deriving the acceptance ratio. Propose $\mathbf{x}' \sim q(\cdot \mid \mathbf{x})$ and accept with probability $\alpha(\mathbf{x}, \mathbf{x}')$, otherwise stay. For $\mathbf{x}' \neq \mathbf{x}$ the kernel is $T(\mathbf{x} \to \mathbf{x}') = q(\mathbf{x}' \mid \mathbf{x})\alpha(\mathbf{x}, \mathbf{x}')$, and detailed balance asks $\pi(\mathbf{x})q(\mathbf{x}' \mid \mathbf{x})\alpha(\mathbf{x}, \mathbf{x}') = \pi(\mathbf{x}')q(\mathbf{x} \mid \mathbf{x}')\alpha(\mathbf{x}', \mathbf{x})$. So $\alpha(\mathbf{x}, \mathbf{x}')/\alpha(\mathbf{x}', \mathbf{x})$ must equal the ratio r below; the largest choice with both values in $[0, 1]$ sets the larger of the two to 1:

$$\alpha(\mathbf{x}, \mathbf{x}') = \min\Big\{1,\ r\Big\}, \qquad r = \frac{\tilde{\pi}(\mathbf{x}')\, q(\mathbf{x} \mid \mathbf{x}')}{\tilde{\pi}(\mathbf{x})\, q(\mathbf{x}' \mid \mathbf{x})}. \tag{11.8}$$

The normalising constant of π cancels. For a symmetric proposal (random walk $\mathbf{x}' = \mathbf{x} + \varepsilon$) the Hastings factor $q(\mathbf{x} \mid \mathbf{x}')/q(\mathbf{x}' \mid \mathbf{x})$ is 1: always accept uphill moves, accept downhill moves with probability $\tilde{\pi}(\mathbf{x}')/\tilde{\pi}(\mathbf{x})$.

Proposition 11.4 (Gibbs is MH with acceptance one). *Resampling coordinate i from its full conditional is an MH step with proposal $q(\mathbf{x}' \mid \mathbf{x}) = \pi(x'_i \mid \mathbf{x}_{-i})\mathbb{1}\{\mathbf{x}'_{-i} = \mathbf{x}_{-i}\}$, and its*

acceptance ratio is identically 1.

Proof. With $\mathbf{x}'_{-i} = \mathbf{x}_{-i}$ and $\pi(\mathbf{x}) = \pi(x_i \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i})$,

$$r = \frac{\pi(x'_i \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i}) \cdot \pi(x_i \mid \mathbf{x}_{-i})}{\pi(x_i \mid \mathbf{x}_{-i})\pi(\mathbf{x}_{-i}) \cdot \pi(x'_i \mid \mathbf{x}_{-i})} = 1. \qquad \square$$

Random-walk MH with step size s: propose $y = x^{(t)} + s\,\varepsilon$, $\varepsilon \sim N(0,1)$, draw $u \sim U(0,1)$, and set $x^{(t+1)} = y$ if $\log u < \log \tilde{\pi}(y) - \log \tilde{\pi}(x^{(t)})$ (work in logs), else $x^{(t+1)} = x^{(t)}$.

Worked Example 11.12 (One Metropolis–Hastings step by hand)**.** Target Gamma$(3,1)$, $\tilde{\pi}(x) = x^2 e^{-x}$ $(x > 0$; the constant $\frac{1}{2}$ cancels). The chain is at $x = 2$ and the proposal is $y = 3.5$. Compute the acceptance probability (a) for a symmetric random-walk proposal, (b) for the independence proposal $q = \text{Exp(mean 3)}$, $q(z) = \frac{1}{3}e^{-z/3}$, which ignores the current state.

Solution. (a) The Hastings factor is 1, so $r = \dfrac{3.5^2 e^{-3.5}}{2^2 e^{-2}} = \dfrac{12.25}{4} e^{-1.5} = 3.0625 \cdot 0.2231 = 0.6833$ and $\alpha = \min\{1, 0.6833\} = 0.6833$. (b) Now $q(\mathbf{x} \mid \mathbf{x}') = q(2)$ and $q(\mathbf{x}' \mid \mathbf{x}) = q(3.5)$, so the Hastings factor is $q(2)/q(3.5) = e^{(3.5-2)/3} = e^{0.5} = 1.6487$, $r = 0.6833 \cdot 1.6487 = 1.1266$ and $\alpha = 1$: the move is accepted for sure, because the proposal is less likely to suggest 3.5 than 2.

PYTHON LAB · `s11_examples.py` · `ex_mh_acceptance()`

```
random walk:  ratio = 0.6833, accept prob = 0.6833 | independence:  Hastings
factor 1.6487, ratio 1.1266, accept 1.0000
```

PART III · MCMC AND BEYOND

11.10 Is the Chain Done? Convergence Diagnostics

MCMC output is a correlated sequence, and no diagnostic can prove convergence; they can only reveal failure.

Definition 11.4 (Autocorrelation time, ESS, $\hat{R}$)**.** For a stationary chain with lag-k autocorrelation ρ_k of $f(\mathbf{x}^{(t)})$, the *integrated autocorrelation time* is $\tau = 1 + 2\sum_{k \geq 1} \rho_k$ and $\text{ESS} = N/\tau$. For M chains of length n with within-chain variance W (mean of the M chain variances, each with divisor $n-1$) and between-chain variance $B = n \cdot \text{Var}(\text{chain means})$ (divisor $M-1$), the Gelman–Rubin statistic is

$$\hat{R} = \sqrt{\widehat{\text{var}}^{+}/W}, \qquad \widehat{\text{var}}^{+} = \tfrac{n-1}{n} W + \tfrac{1}{n} B. \tag{11.9}$$

Here $\widehat{\text{var}}^{+}$ estimates the target variance from all chains together; if the chains have not mixed, the between-chain spread B inflates it relative to W and $\hat{R} > 1$. In practice the sum defining τ is truncated at the first lag where the (estimated) autocorrelations become negative (Geyer's initial-positive-sequence rule, used by `s11_examples.py`).

In practice: inspect trace plots for trends, stuck chains and rare mode switches; discard a burn-in; thin only to save memory (thinning never increases the ESS of all draws); report the ESS, not N (an ESS of a few hundred suffices for two-digit summaries); and run several dispersed chains, requiring $\hat{R} < 1.01$ — values above 1.1 mean the chains disagree.

Worked Example 11.13 ($\hat{R}$ by hand)**.** Three chains of length $n = 4$ (after burn-in) are $(1.0, 1.4, 0.6, 1.0)$, $(1.2, 0.8, 1.0, 1.4)$ and $(2.2, 2.6, 1.8, 2.2)$. Compute $\hat{R}$.
Solution. Chain means 1.0, 1.1, 2.2 (grand mean 1.4333). Chain variances (divisor $n-1=3$): $(0+0.16+0.16+0)/3 = 0.1067$, $(0.01+0.09+0.01+0.09)/3 = 0.0667$ and 0.1067; so $W = (0.1067+0.0667+0.1067)/3 = 0.0933$. Variance of the chain means (divisor $M-1=2$): $\big((-0.4333)^2+(-0.3333)^2+0.7667^2\big)/2 = 0.8867/2 = 0.4433$, so $B = 4\cdot 0.4433 = 1.7733$. Then $\widehat{\mathrm{var}}^{+} = \frac{3}{4}\cdot 0.0933 + \frac{1}{4}\cdot 1.7733 = 0.0700 + 0.4433 = 0.5133$ and $\hat{R} = \sqrt{0.5133/0.0933} = 2.345$: the third chain is stuck elsewhere. Dropping it gives $\hat{R} = 0.899$ (values slightly below 1 occur for very short chains).

PYTHON LAB · `s11_examples.py` · `ex_rhat_hand()`

```
chain means [1.  1.1 2.2]; W = 0.0933, B = 1.7733, var+ = 0.5133, R-hat =
2.345; without chain 3:  R-hat = 0.899
```

Worked Example 11.14 (Step size on a bimodal target)**.** Run random-walk MH on the illustrative mixture $\pi = 0.3\,N(-3, 0.8^2) + 0.7\,N(2, 1)$ (mean $0.3(-3)+0.7(2) = 0.5$) with step sizes $s = 0.2$, 2.5 and 25; four chains of 5000 iterations from $-4, -3, 4, 5$ (seed 2026); discard 500 as burn-in, leaving $4\cdot 4500 = 18{,}000$ draws. Report the acceptance rate, the pooled mean, the total ESS (sum of the four per-chain ESS of Definition 11.4) and $\hat{R}$ (11.9).
Solution (Fig. 11.6; simulation output). Step 0.2 accepts 93% of proposals but never crosses between the modes: each chain stays in the mode where it started (two near -3, two near 2), so the pooled mean is $-0.52 \approx \frac{1}{4}(-3-3+2+2) = -0.5$ instead of 0.5; the total ESS is 20 out of 18,000 draws and $\hat{R} = 1.23$ flags the failure. Step 2.5 accepts 48% and mixes well (ESS 1167, $\hat{R} = 1.00$, mean 0.56). Step 25 accepts only 9% — most proposals land in the tails — and the ESS drops to 706 ($\hat{R} = 1.00$, mean 0.38). High acceptance is *not* a sign of a good sampler (for random-walk MH in high dimension the optimal acceptance rate is about 0.234; Roberts, Gelman & Gilks 1997).

PYTHON LAB · `s11_examples.py` · `ex_mh_random_walk()`

```
step 0.2:  acc 0.93, mean -0.52, ESS 20, R-hat 1.23 | 2.5:  0.48, +0.56,
1167, 1.00 | 25:  0.09, +0.38, 706, 1.00
```

Worked Example 11.15 (Autocorrelation of Gibbs on a correlated Gaussian)**.** Run Gibbs on $(X, Y) \sim N\big(0, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)\big)$ using $X \mid Y = y \sim N(\rho y, 1-\rho^2)$ and symmetrically for Y; $N = 20{,}000$ sweeps from $(0, 0)$ (seed 2026), for $\rho = 0.5$ and $\rho = 0.95$.
Solution. By Proposition 11.7 the X-chain is AR(1) with coefficient ρ^2, so $\tau = (1+\rho^2)/(1-\rho^2)$ and the theoretical ESS is N/τ. For $\rho = 0.5$: $\rho^2 = 0.25$, $\tau = 1.25/0.75 = 1.67$ and $N/\tau = 12{,}000$; the simulated chain has lag-1 autocorrelation 0.247 and estimated ESS 12,378. For $\rho = 0.95$: $\rho^2 = 0.9025$, $\tau = 1.9025/0.0975 = 19.51$ and $N/\tau = 1025$; simulated lag-1 autocorrelation 0.901, estimated ESS 1098 — 20,000 sweeps are worth about a thousand independent draws (Fig. 11.5, right).

PYTHON LAB · `s11_examples.py` · `ex_gibbs_bvn()`

```
rho = 0.5:  lag-1 = 0.247 (0.2500), tau = 1.67, ESS 12378 (theory 12000) |
rho = 0.95:  lag-1 = 0.901, tau = 19.51, ESS 1098 (1025)
```

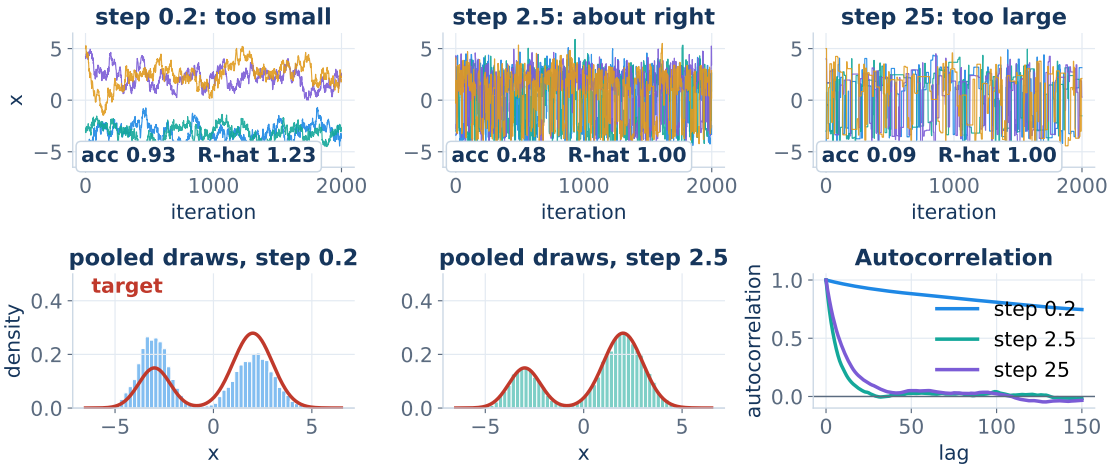


Figure 11.6. Random-walk MH on the bimodal target of Worked Example 11.14. Top: the first 2000 iterations of the four chains for each step size, with acceptance rate and $\hat{R}$. Bottom: pooled post-burn-in draws against the target, and the autocorrelation functions. Too small a step gives high acceptance but no mixing between modes; too large a step rejects almost everything.

PART III · MCMC AND BEYOND

11.11 Variational Inference in One Section

Variational inference (VI) replaces sampling by optimisation. For latent $\mathbf{z}$, data $\mathbf{x}$ and any density $q(\mathbf{z})$,

$$\log p(\mathbf{x}) = \underbrace{\mathbb{E}_q[\log p(\mathbf{x},\mathbf{z})] - \mathbb{E}_q[\log q(\mathbf{z})]}_{\mathrm{ELBO}(q)} + \mathrm{KL}\big(q(\mathbf{z}) \,\|\, p(\mathbf{z}\mid\mathbf{x})\big), \tag{11.10}$$

so maximising the ELBO over a tractable family is the same as minimising $\mathrm{KL}(q\|p(\cdot\mid\mathbf{x}))$; the ELBO is the bound EM maximised in Session 10. In *mean field*, $q(\mathbf{z}) = \prod_j q_j(z_j)$, and coordinate ascent (CAVI) iterates

$$\log q_j^*(z_j) = \mathbb{E}_{q_{-j}}\big[\log p(\mathbf{x},\mathbf{z})\big] + \text{const}, \tag{11.11}$$

a "Gibbs sampler with expectations in place of samples" that needs only each variable's Markov blanket. For a bivariate Gaussian target $N(\boldsymbol{\mu},\Sigma)$ with precision $\Lambda = \Sigma^{-1}$, $\log p = -\tfrac{1}{2}\Lambda_{11}(z_1-\mu_1)^2 - \Lambda_{12}(z_1-\mu_1)(z_2-\mu_2) + (\text{terms without } z_1)$; taking $\mathbb{E}_{q_2}$ replaces z_2 by its mean m_2, and completing the square in z_1, (11.11) gives $q_1 = N(m_1, 1/\Lambda_{11})$ with $m_1 = \mu_1 - \Lambda_{12}(m_2-\mu_2)/\Lambda_{11}$ (and symmetrically for q_2).

Example (Fig. 11.7, left). Target $N\big((1,-1),\Sigma\big)$ with unit variances and correlation $\rho = 0.9$. Then $\Lambda = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$, so $\Lambda_{11} = \Lambda_{22} = 1/(1-0.81) = 1/0.19 = 5.26$ and $-\Lambda_{12}/\Lambda_{11} = \rho = 0.9$. Each factor has variance $1/5.26 = 0.19$ against a true marginal variance of 1 (Proposition 11.8). At the optimum the means are exact, and the Gaussian KL formula gives $\mathrm{KL}(q^*\|p) = \tfrac{1}{2}\big[\mathrm{tr}(\Lambda S_q) - 2 + \ln(\det\Sigma/\det S_q)\big]$ with $S_q = \mathrm{diag}(0.19, 0.19)$: $\mathrm{tr}(\Lambda S_q) = 2$ and $\det\Sigma/\det S_q = (1-\rho^2)/(1-\rho^2)^2$, so $\mathrm{KL}(q^*\|p) = -\tfrac{1}{2}\ln(1-\rho^2) = -\tfrac{1}{2}\ln 0.19 = 0.8304$. CAVI from $(0,0)$ updates $m_1 = 1 + 0.9(m_2+1)$, then $m_2 = -1 + 0.9(m_1 - 1)$: sweep 1 gives $m_1 = 1 + 0.9\cdot 1 = 1.90$ and $m_2 = -1 + 0.9\cdot 0.90 = -0.19$; then $(1.73, -0.34)$, …, $(1.39, -0.65)$ after five sweeps. The means converge to $(1,-1)$, but the error $m_1 - 1$ only shrinks by the factor $\rho^2 = 0.81$ per sweep $(0.90, 0.729, \dots)$ — the same zig-zag as Gibbs.

Which KL? (Fig. 11.7, right). Fit a single Gaussian $q = N(m, s^2)$ to the bimodal target of Worked Example 11.14. Minimising $\mathrm{KL}(p\|q)$ matches moments: $m = 0.5$ and $s^2 = 0.3(0.64 +$

$9) + 0.7(1 + 4) - 0.5^2 = 6.142$, $s = 2.478$, a broad q covering both modes. Minimising $\mathrm{KL}(q\|p)$ (the VI objective; numerical grid search over m, s in steps of 0.05) gives $m = 1.950$, $s = 1.050$ with $\mathrm{KL} = 0.350$: q locks onto the heavier mode. For comparison, $q = N(2, 1)$ exactly would give $\mathrm{KL}(q\|p) \approx -\ln 0.7 = 0.357$, because under that q the target density is about $0.7\,q$.

PYTHON LAB · `s11_examples.py` · `ex_mean_field_gaussian()`, `ex_kl_asymmetry()`

```
mean-field var 0.19 (true 1); KL(q*||p) = 0.8304; sweep 1:  (1.9000,
-0.1900); sweep 5:  (1.3874, -0.6513)
KL(p||q) fit:  mean 0.500, sd 2.478 (var 6.142) | KL(q||p) fit:  mean 1.950,
sd 1.050, KL 0.350; -ln 0.7 = 0.357
```

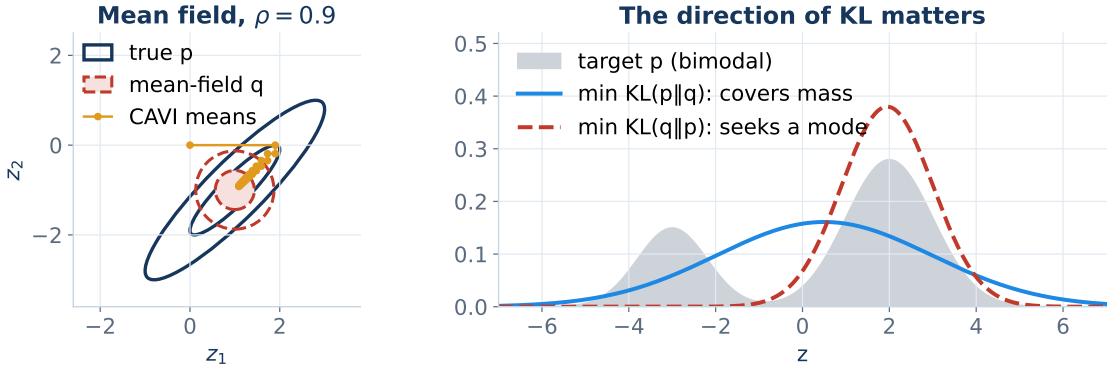


Figure 11.7. Left: mean-field VI fits a too-narrow, uncorrelated q (dashed circles) to a correlated Gaussian; amber: the CAVI path of the means. Right: Gaussian fits to the bimodal target of Worked Example 11.14. $\mathrm{KL}(p\|q)$ matches moments ($N(0.5, 2.478^2)$) and spreads over both modes; $\mathrm{KL}(q\|p)$, the VI objective, locks onto the heavier mode ($N(1.95, 1.05^2)$, $\mathrm{KL} = 0.350$, just below $-\ln 0.7 = 0.357$).

AI CONNECTION · Generative Models

A variational autoencoder (Kingma & Welling 2014; course session 4) maximises the ELBO (11.10) with an *amortised* $q_\phi(\mathbf{z} \mid \mathbf{x})$: an encoder outputs the variational parameters for every $\mathbf{x}$ instead of running CAVI per data point; diffusion models also train on an ELBO. $\mathrm{KL}(q\|p)$ is infinite unless $q = 0$ wherever $p = 0$ (zero-forcing, mode-seeking), whereas $\mathrm{KL}(p\|q)$ punishes missing mass and matches moments (Fig. 11.7): hence over-confident VI and VAE posteriors.

PART III · MCMC AND BEYOND

11.12 Particle Filtering

For dynamic models (HMMs and dynamic BNs with continuous or huge state spaces), the filtering recursion of Session 7, $b_t(x_t) \propto p(o_t \mid x_t) \int p(x_t \mid x_{t-1}) b_{t-1}(x_{t-1})\, dx_{t-1}$, has no closed form outside the linear-Gaussian (Kalman) case. A *particle filter* (bootstrap filter; Gordon, Salmond & Smith 1993) represents b_t by N weighted samples and runs sequential importance sampling with the transition model as proposal: (1) *propagate* $x_t^{(k)} \sim p(x_t \mid x_{t-1}^{(k)})$; (2) *weight* $w_t^{(k)} \propto w_{t-1}^{(k)} p(o_t \mid x_t^{(k)})$ and normalise; (3) *resample* N particles with probabilities $w_t^{(k)}$ (e.g. systematic resampling) when the ESS (11.7) drops below $N/2$, and reset all weights to $1/N$. Without step 3 the weights degenerate: after a few steps one particle carries almost all the weight.

Worked Example 11.16 (A robot in a corridor)**.** A robot starts at $x_0 = 1.5$ and moves along a corridor $[0, 20]$ with doors at $3, 7, 14$: $x_t = x_{t-1} + 1 + \varepsilon_t$ with noise s.d. 0.2 (the particles are propagated with a slightly larger s.d. 0.3). Its door sensor reports $o_t = 1$ w.p. 0.9 when the robot is within 0.6 of a door and w.p. 0.1 elsewhere, so a particle's likelihood is 0.9 if its own door status agrees with o_t and 0.1 otherwise (particles beyond 20 get likelihood 10^{-6}). Start from a uniform belief on $[0, 20]$ with $N = 1000$ particles and filter for $t = 0, \ldots, 14$ (simulation, seed 2030).
Solution (simulation output). After the first (no-door) reading the belief is still nearly uniform (mean 10.42, s.d. 5.81; a uniform law on $[0, 20]$ has mean 10, s.d. 5.77). The first door reading ($t = 1$) up-weights the particles near the three doors by a factor $0.9/0.1 = 9$ and the ESS falls to $316.7 < 500$, triggering resampling; the second door ($t = 5$) favours hypotheses whose door *spacing* matches; after the third ($t = 12$) the belief is concentrated, mean 13.97, s.d. 0.63, against the true 14.12, and at $t = 14$ it is 16.10, s.d. 0.51 (truth 16.47). Resampling occurred at 3 of the 15 steps ($t = 1, 5, 11$).

PYTHON LAB · `s11_examples.py` · `ex_particle_filter()`

```
t=1:  ESS = 316.7 -> resample; t=12:  true 14.12, mean 13.97, sd 0.63; t=14:
16.47 vs 16.10 (sd 0.51); 3 resamplings
```

Table 11.2. The approximate-inference toolbox of this session.

Method	Needs / output	Fails when
Rejection	ancestral sampler; exact draws from $P(\cdot \mid e)$	$P(e)$ tiny
Likelihood weighting	CPTs of E; weighted draws	evidence at the leaves, ESS collapses
Importance sampling	p up to a constant, proposal q	q lighter-tailed than $\lvert f\rvert p$
Gibbs	samplable full conditionals; correlated draws	strong coupling, separated modes
Metropolis–Hastings	$\tilde{\pi}$ and any proposal; correlated draws	badly tuned step, separated modes
Mean-field VI	expectations of $\log p$; fast, deterministic, biased	strong correlations (variance too small)
Particle filter	transition sampler, likelihood; sequential	very informative sensors, high dimension

PART IV

The Statistics of Sampling and Model Selection

Unbiasedness, variance, optimal proposals, Markov-chain CLTs and BIC asymptotics

PART IV · MATHEMATICAL STATISTICS

11.13 Monte Carlo Estimators

Theorem 11.2 (Monte Carlo estimator)**.** *Let* $\mathbf{X}^{(1)}, \dots, \mathbf{X}^{(N)} \overset{iid}{\sim} P$ *and* f *with* $\mathbb{E}f^2 < \infty$, $\mu = \mathbb{E}f$, $\sigma^2 = \mathrm{Var}f$. *Then* $\hat{\mu}_N$ *of* (11.4) *satisfies (a)* $\mathbb{E}\hat{\mu}_N = \mu$; *(b)* $\mathrm{Var}\hat{\mu}_N = \sigma^2/N$; *(c)* $\hat{\mu}_N \to \mu$ *almost surely; (d)* $\sqrt{N}(\hat{\mu}_N - \mu) \xrightarrow{d} N(0, \sigma^2)$, *and with the sample s.d.* s_N, $\hat{\mu}_N \pm z_{1-\alpha/2}\, s_N/\sqrt{N}$ *is an asymptotic* $1-\alpha$ *confidence interval.*

Proof. (a) Linearity of expectation. (b) The $f(\mathbf{X}^{(m)})$ are independent, so variances add: $\mathrm{Var}\hat{\mu}_N = N^{-2} \cdot N\sigma^2$. (c) Strong law of large numbers. (d) The Lindeberg–Lévy CLT gives the limit; $s_N \xrightarrow{p} \sigma$ by the law of large numbers applied to f and f^2 and the continuous mapping theorem, so by Slutsky's theorem $\sqrt{N}(\hat{\mu}_N - \mu)/s_N \xrightarrow{d} N(0,1)$, which is the interval. □

MATHEMATICAL STATISTICS · How many samples? Dimension-free, but not free

The rate $\sigma/\sqrt{N}$ does not depend on the dimension of $\mathbf{x}$, but σ can be huge: estimating a small probability p with relative error $\mathrm{SE}/p = \varepsilon$ needs $\sqrt{p(1-p)/N} = \varepsilon p$, i.e. $N = (1-p)/(p\varepsilon^2)$: $0.999/(0.001 \cdot 0.01) = 99{,}900 \approx 10^5$ draws for $p = 10^{-3}$, $\varepsilon = 0.1$. For a Bernoulli query $\sigma^2 = p(1-p) \le \frac{1}{4}$, so the required $N = \sigma^2/\mathrm{SE}^2$ is at most $1/(4\,\mathrm{SE}^2)$ (10,000 for $\mathrm{SE} = 0.005$; Worked Example 11.6 needed 9135). Variance reduction (importance sampling, control variates, Rao–Blackwellisation) attacks σ, not $\sqrt{N}$.

Theorem 11.3 (Importance sampling: variance and optimal proposal)**.** *Let* $q > 0$ *wherever* $fp \neq 0$. *Then* $\hat{\mu}_{\mathrm{IS}}$ *is unbiased with*

$$\mathrm{Var}_q(\hat{\mu}_{\mathrm{IS}}) = \frac{1}{N}\Big(\int \frac{f^2 p^2}{q}\, dx - \mu^2\Big), \tag{11.12}$$

which is minimised over densities q *by* $q^*(x) = |f(x)|p(x)/\int |f|p$, *with minimum* $\frac{1}{N}\big[(\int |f|p)^2 - \mu^2\big]$. *If* $f \ge 0$, q^* *has variance zero.*

Proof. Unbiasedness is (11.6): $\mathbb{E}_q[wf] = \int q \frac{p}{q} f = \mu$. The variance of one term is $\mathbb{E}_q[w^2 f^2] - \mu^2 = \int f^2 p^2/q - \mu^2$. By Cauchy–Schwarz, for any density q,

$$\Big(\int |f|p\Big)^2 = \Big(\int \frac{|f|p}{\sqrt{q}}\sqrt{q}\Big)^2 \le \int \frac{f^2p^2}{q} \int q = \int \frac{f^2p^2}{q},$$

with equality iff $|f|p/\sqrt{q} \propto \sqrt{q}$, i.e. $q \propto |f|p$. For $f \ge 0$, $(\int fp)^2 = \mu^2$, so the minimum variance is 0: every weighted sample $wf = \int fp$ equals μ exactly. □

q^* needs the unknown μ, but says what to aim for: sample where $|f|p$ is large, and never let q have lighter tails than $|f|p$ (else $\int f^2p^2/q = \infty$).

Proposition 11.5 (ESS under a shifted proposal)**.** *For target* $p = N(0,1)$ *and proposal* $q = N(\mu, 1)$, $w(y) = e^{-\mu y + \mu^2/2}$, $\mathbb{E}_q[w^2] = e^{\mu^2}$, *and* $\mathrm{ESS}/N \to e^{-\mu^2}$. *For the tail event* $f = \mathbb{1}\{y > c\}$, *the per-sample variance of* $\hat{\mu}_{\mathrm{IS}}$ *is* $e^{\mu^2}\bar{\Phi}(c+\mu) - \bar{\Phi}(c)^2$.

Proof. $\mathbb{E}_q[w^2] = \int p^2/q$ and $\phi(y)^2/\phi(y-\mu) = \frac{1}{\sqrt{2\pi}}\exp\{-y^2 + \frac{1}{2}(y-\mu)^2\} = \phi(y+\mu)e^{\mu^2}$, which integrates to e^{μ^2}; restricting to $y > c$ gives $e^{\mu^2}\bar{\Phi}(c+\mu)$. Since $\mathbb{E}_q w = 1$, the law of large numbers gives $\mathrm{ESS}/N = (\bar{w})^2/\overline{w^2} \to 1/\mathbb{E}_q[w^2]$. □

A shift of two standard deviations already leaves $e^{-4} \approx 1.8\%$ of the samples effective. For a target and proposal on n independent coordinates, each shifted by μ, the weight is a product of n such factors, so $\mathbb{E}_q[w^2] = e^{n\mu^2}$ and $\mathrm{ESS}/N \to e^{-n\mu^2}$: the curse of dimensionality of importance sampling, and the reason particle filters must resample.

PART IV · MATHEMATICAL STATISTICS

11.14 The Statistics of MCMC

Proposition 11.6 (Detailed balance and the MH kernel)**.** *(a) If T satisfies detailed balance with respect to π, then π is stationary for T. (b) The MH kernel with acceptance (11.8) satisfies detailed balance with respect to π.*

Proof. (a) $\sum_{\mathbf{x}} \pi(\mathbf{x})T(\mathbf{x}\to\mathbf{x}') = \sum_{\mathbf{x}} \pi(\mathbf{x}')T(\mathbf{x}'\to\mathbf{x}) = \pi(\mathbf{x}')\sum_{\mathbf{x}} T(\mathbf{x}'\to\mathbf{x}) = \pi(\mathbf{x}')$, since each row of T sums to one. (b) For $\mathbf{x}' \neq \mathbf{x}$,

$$\pi(\mathbf{x})T(\mathbf{x}\to\mathbf{x}') = \pi(\mathbf{x})q(\mathbf{x}' \mid \mathbf{x})\min\{1, r\} = \min\big\{\pi(\mathbf{x})q(\mathbf{x}' \mid \mathbf{x}),\ \pi(\mathbf{x}')q(\mathbf{x} \mid \mathbf{x}')\big\},$$

which is symmetric in $\mathbf{x}, \mathbf{x}'$; for $\mathbf{x}' = \mathbf{x}$ detailed balance is trivial. (Continuous states: replace sums by integrals.) □

MATHEMATICAL STATISTICS · Ergodic theorem and the Markov-chain CLT

Ergodic theorem. If a chain on a finite state space is irreducible and aperiodic with stationary π, then π is unique, $P(\mathbf{X}^{(t)} = \cdot \mid \mathbf{X}^{(0)} = \mathbf{x}) \to \pi$ for every start $\mathbf{x}$, and $\frac{1}{N}\sum_t f(\mathbf{X}^{(t)}) \to \mathbb{E}_\pi f$ almost surely (general state spaces: Harris recurrence). **CLT.** Under geometric ergodicity and $\mathbb{E}_\pi|f|^{2+\delta} < \infty$, $\sqrt{N}(\hat{\mu}_N - \mu) \xrightarrow{d} N(0, \sigma^2\tau)$ with $\tau = 1 + 2\sum_{k\geq 1}\rho_k$: the chain behaves like N/τ independent draws. Gibbs on a positive Markov network is irreducible and aperiodic; random-walk MH is too whenever $\pi > 0$ everywhere.

Proposition 11.7 (Gibbs on a bivariate normal is AR(1))**.** *For the Gibbs sampler of Worked Example 11.15, $X_t = \rho^2 X_{t-1} + \eta_t$ with $\eta_t \overset{iid}{\sim} N(0, 1-\rho^4)$. Hence $\rho_k = \rho^{2k}$ and $\tau = (1+\rho^2)/(1-\rho^2)$.*

Proof. $Y_{t-1} = \rho X_{t-1} + \sqrt{1-\rho^2}\,\varepsilon'_{t-1}$ and $X_t = \rho Y_{t-1} + \sqrt{1-\rho^2}\,\varepsilon_t$ with independent standard normals, so $X_t = \rho^2 X_{t-1} + \eta_t$, $\eta_t = \rho\sqrt{1-\rho^2}\,\varepsilon'_{t-1} + \sqrt{1-\rho^2}\,\varepsilon_t$, independent of the past with variance $(1-\rho^2)(\rho^2+1) = 1-\rho^4$. For an AR(1) with coefficient $a = \rho^2$, $\rho_k = a^k$ and $\tau = 1 + 2\sum_{k\geq 1} a^k = (1+a)/(1-a)$. □

As $\rho \to 1$, $\tau \to \infty$: coordinate-wise samplers (and CAVI) crawl on strongly coupled variables; block or reparametrise them.

Proposition 11.8 (Mean-field variances are too small)**.** *For a Gaussian target, the optimal mean-field factor of z_i has variance $1/\Lambda_{ii} = \mathrm{Var}(z_i \mid \mathbf{z}_{-i}) \le \mathrm{Var}(z_i) = \Sigma_{ii}$, with equality iff z_i is uncorrelated with $\mathbf{z}_{-i}$.*

Proof. From (11.11), $\log q_i^*(z_i) = -\tfrac{1}{2}\Lambda_{ii} z_i^2 + (\text{linear in } z_i)$, a normal with variance $1/\Lambda_{ii}$, which is the conditional variance by the precision form of Gaussian conditioning (Session 5 appendix; Session 7). By the law of total variance, $\mathrm{Var} z_i = \mathbb{E}\mathrm{Var}(z_i \mid \mathbf{z}_{-i}) + \mathrm{Var}\mathbb{E}(z_i \mid \mathbf{z}_{-i}) \ge \mathrm{Var}(z_i \mid \mathbf{z}_{-i})$, with equality iff the regression of z_i on $\mathbf{z}_{-i}$ is flat. □

PART IV · MATHEMATICAL STATISTICS

11.15 Asymptotics of Model Selection

MATHEMATICAL STATISTICS · BIC as a Laplace approximation

Write $P(\mathcal{D} \mid \mathcal{G}) = \int e^{\ell(\theta)} p(\theta)\, d\theta$. Around the MLE, $\ell(\theta) \approx \ell(\hat\theta) - \frac{N}{2}(\theta - \hat\theta)^\top \hat I_1 (\theta - \hat\theta)$ with $\hat I_1$ the per-observation observed information, so a Gaussian integral gives

$$\log P(\mathcal{D} \mid \mathcal{G}) = \ell(\hat\theta) - \frac{d}{2}\log N + \underbrace{\log p(\hat\theta) + \tfrac{d}{2}\log 2\pi - \tfrac{1}{2}\log\det \hat I_1}_{O(1)} + o(1).$$

Dropping the $O(1)$ terms is BIC (Schwarz 1978). In Worked Example 11.2 the log Bayes factor 16.73 and the BIC difference 16.25 differ by exactly such an $O(1)$ term, 0.48.

Proposition 11.9 (The BIC edge test)**.** *For two discrete variables with r_A and r_B values, $\mathrm{BIC}(A \to B) - \mathrm{BIC}(A, B \text{ independent}) = \tfrac{1}{2}\big(G^2 - \mathrm{df}\log N\big)$, where $G^2 = 2N\hat I(A;B)$ is the likelihood-ratio statistic and $\mathrm{df} = (r_A - 1)(r_B - 1)$. AIC adds the edge iff $G^2 > 2\,\mathrm{df}$. Consequently, if $A \perp\!\!\!\perp B$, $P(\text{BIC adds the edge}) \to 0$ while $P(\text{AIC adds it}) \to P(\chi^2_{\mathrm{df}} > 2\,\mathrm{df}) > 0$; if $A \not\perp\!\!\!\perp B$ both add it with probability $\to 1$.*

Proof. By (11.1) the two log-likelihoods differ by $N\hat I(A;B) = G^2/2$, and the parameter counts by $(r_B - 1)r_A - (r_B - 1) = (r_A - 1)(r_B - 1)$ (the B-family has r_A rows of $r_B - 1$ free entries with the edge, one row without it; the A-family is shared). Substituting into (11.2) gives both thresholds. Under independence, Wilks' theorem gives $G^2 \xrightarrow{d} \chi^2_{\mathrm{df}}$, so $P(G^2 > \mathrm{df}\log N) \to 0$ because $\log N \to \infty$, while the AIC threshold stays fixed. Under dependence, $G^2/N \to 2I(A;B) > 0$ almost surely, so G^2 eventually exceeds $\mathrm{df}\log N = o(N)$. □

For binary A, B the AIC false-edge rate tends to $P(\chi^2_1 > 2) = 0.157$. A simulation makes this concrete: 4000 simulated data sets per N with $P(A{=}1) = 0.5$ and either no edge ($P(B{=}1 \mid a) = 0.5$ for both a) or a weak edge ($P(B{=}1 \mid A{=}1) = 0.58$, $P(B{=}1 \mid A{=}0) = 0.42$). Without an edge, BIC adds one in 5.2% of the data sets at $N = 50$ (the χ^2_1 approximation predicts $P(\chi^2_1 > \ln 50) = 4.8\%$) and in 0.1% at $N = 20{,}000$, while AIC adds one in 13.9%–16.4% at every N. The weak true edge is found by BIC in 19.8% ($N = 50$), 49.0% ($N = 200$) and 99.3% ($N = 1000$) of the data sets, by AIC earlier (39.4%, 80.7%, 100%). BIC is *consistent*; AIC is *efficient* for prediction (`ex_bic_consistency()`).

PART IV · MATHEMATICAL STATISTICS

11.16 Worked Examples

Worked Example 11.17 (Estimating a rare-evidence query by ancestral sampling)**.** Suppose that in some version of the student network $P(L = \text{strong})$ is rare (hypothetically, true probability 0.02). To estimate $P(I = \text{high} \mid L = \text{strong})$ by naive rejection sampling from ancestral draws, we would expect to need roughly $1/0.02 = 50$ full samples on average just to see *one* usable draw matching the evidence — for a query on a rarer event (say probability 0.0002), that becomes 5,000 wasted samples per usable one. This is exactly the motivation for importance sampling (reweight instead of reject) and, in harder cases with no convenient proposal, for MCMC methods like Gibbs sampling that never need to "match" evidence by chance — evidence is instead fixed once and the sampler only ever resamples the unobserved variables' full conditionals.

Additional Worked Examples (Instructor Supplement)

Worked Example 11.18 (Rejection sampling can fail even with "enough" draws on average)**.** If rare evidence has true probability $p = 0.02$, the expected number of ancestral draws needed for one accepted sample is $1/p = 50$ (as noted above). Suppose we budget exactly 100 draws (twice the expectation). What is the probability we get *zero* usable samples?
Solution. Each draw independently matches the evidence with probability $p = 0.02$, so the number of matches in 100 draws is Binomial$(100, 0.02)$, and

$$P(\text{zero matches in 100 draws}) = (1 - 0.02)^{100} = 0.98^{100}.$$

Using $\ln(0.98) \approx -0.020203$, we get $100 \ln(0.98) \approx -2.0203$, so $0.98^{100} = e^{-2.0203} = \mathbf{0.1326}$. Even at *twice* the expected number of draws, there is still roughly a 13% chance of walking away with no usable sample at all — a stark, quantitative reason rejection sampling is considered impractical for rare evidence, well beyond the qualitative "you'll waste a lot of samples" framing, and a concrete motivation for both importance sampling (reweight instead of discard) and MCMC (never needs to match evidence by chance in the first place).

PYTHON LAB · `s11_examples.py` · `ex_rare_evidence()`

```
p = 0.02:  50 draws per accepted sample; p = 0.0002:  5,000; P(zero accepted
in 100 draws) = 0.1326
```

(The supplement's BIC calculation is Worked Example 11.1; its full conditional is Worked Example 11.11.)

New Numerical and Analytical Examples

Worked Example 11.19 (Markov-equivalent DAGs have equal likelihood (analytical))**.** Show that $A \to B$ and $B \to A$ have the same maximised likelihood for any complete data.
Solution. By (11.1), $\ell(A \to B) = N[\hat{I}(B; A) - \hat{H}(A) - \hat{H}(B)]$ and $\ell(B \to A) = N[\hat{I}(A; B) - \hat{H}(A) - \hat{H}(B)]$ (the parentless node contributes $\hat{I} = 0$); mutual information is symmetric.

More generally, the maximised likelihood of a DAG is $N \sum_{\mathbf{x}} \hat{P}(\mathbf{x}) \log \tilde{P}(\mathbf{x})$, where $\tilde{P}$ is the best-fitting distribution among those that factorise over the DAG. Markov-equivalent DAGs (same skeleton and v-structures, Theorem 11.1) represent exactly the same set of distributions, hence have the same $\tilde{P}$ and the same likelihood; since they also have the same number of parameters (Chickering 1995), AIC and BIC are score equivalent. Worked Example 11.1 confirms -117.34 both ways.

Worked Example 11.20 (Why rejection is a special case of importance sampling (analytical)). Show that rejection sampling for $P(y \mid e)$ is self-normalised IS with proposal $Q = P$ and weight $w(\mathbf{x}) = \mathbb{1}\{\mathbf{x}_E = e\}$, and that its ESS equals the number of accepted samples.
Solution. The target is $P(\mathbf{x} \mid e) = P(\mathbf{x})\mathbb{1}\{\mathbf{x}_E = e\}/P(e)$, so $p/q \propto \mathbb{1}\{\mathbf{x}_E = e\}$ and (11.6) becomes $\sum_m \mathbb{1}\{e^{(m)}\}\mathbb{1}\{y^{(m)} = y\}/\sum_m \mathbb{1}\{e^{(m)}\}$, the fraction of accepted samples with $Y = y$. With 0/1 weights, $(\sum w)^2/\sum w^2 = K^2/K = K$, the number accepted. LW replaces the hard 0/1 weight by the soft likelihood $P(e \mid \mathrm{pa})$, which is why its ESS (7012 in Worked Example 11.7) is far larger than rejection's 2852.

PART V

Sampling and Structure in Agents

Rollouts, beliefs, votes, world models and interaction graphs

PART V · AI CONNECTIONS

11.17 Sampling Inside Agents

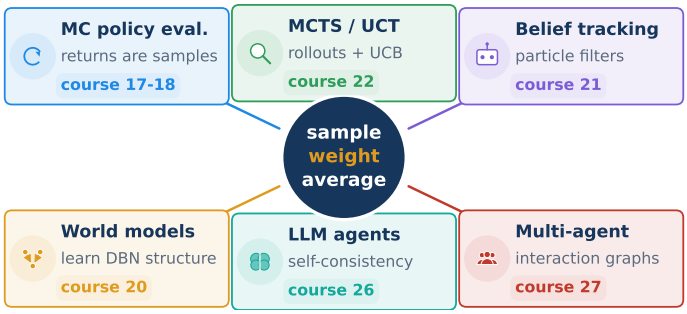


Figure 11.8. Sample, weight, average: where today's estimators reappear in the course.

AI CONNECTION · Reinforcement Learning

Monte Carlo policy evaluation estimates $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ by averaging sampled returns (Theorem 11.2). On the 5-state random walk of Sutton & Barto (2018, Example 6.2) — states $A, \ldots, E$, start in C, step left or right with probability $\frac{1}{2}$, reward 1 for exiting on the right and 0 otherwise, no discounting — $V(C) = 0.5$. 100 simulated episodes (seed 2028) give $\hat{V}(C) = 0.550$, SE $= s/\sqrt{100} = 0.050$, 95% CI $0.550 \pm 1.96 \cdot 0.050 = [0.452, 0.648]$, mean episode length 8.50 (theory $3 \cdot 3 = 9$, the gambler's-ruin duration from distance 3 to each exit); SE $= 0.01$ would need $\sigma^2/0.01^2 = 0.25/0.0001 = 2500$ episodes, the variance problem

that temporal-difference learning attacks by bootstrapping (`ex_mc_policy_eval()`).

LINK TO COURSE SESSION 17 · RL and MDPs (and 18: Dynamic Programming & Q-Learning)

First-visit MC, off-policy MC with weights $\prod_t \pi(a_t \mid s_t)/b(a_t \mid s_t)$ (variance exploding with the horizon, as in Proposition 11.5) and TD learning are this Part's estimators with returns in place of f.

LINK TO COURSE SESSION 22 · Advanced Search and Planning

Monte Carlo tree search scores moves by random rollouts and expands the child maximising $\bar{X}_j + c\sqrt{\ln n / n_j}$ (UCT; Kocsis & Szepesvári 2006): a sample mean plus a bonus shrinking like the standard error of Theorem 11.2. AlphaGo (Silver et al. 2016) combined MCTS with learned policy and value networks.

LINK TO COURSE SESSION 21 · Decision Making under Uncertainty

When a POMDP's state space is continuous or huge, the belief b_t is a particle filter (Worked Example 11.16): Monte Carlo localisation (Thrun, Burgard & Fox 2005) and particle beliefs in online planners.

AI CONNECTION · Agentic AI

Self-consistency (Wang et al. 2023) samples N chain-of-thought paths and returns the plurality answer, a Monte Carlo estimate of the mode of the answer distribution. Simulation (200,000 hypothetical questions per N, ties broken at random): for a model right 40% of the time with errors spread over four wrong answers (15% each), the accuracy of the plurality vote rises from 0.40 ($N = 1$) to 0.55 ($N = 5$), 0.71 ($N = 11$) and 0.96 ($N = 41$). If one wrong answer is more likely than the right one (45%, the other three 5% each), voting *converges to the wrong answer*: 0.36 at $N = 41$ (Fig. 11.9, left). Sampling reduces variance, never bias; best-of-N and tree-of-thought rollouts (course session 26) share this caveat (`ex_self_consistency()`).

PART V · AI CONNECTIONS

11.18 Learning Structure in Agents

Worked Example 11.21 (A factored world model for model-based RL)**.** Hypothetical binary state factors x_1, x_2, x_3 and binary action a, all drawn uniformly at random, with true dynamics $P(x_1'{=}1) = 0.9$ if $a = 1$, else 0.7 if $x_1 = 1$ and 0.1 if $x_1 = 0$; $P(x_2'{=}1 \mid x_1, x_2) = 0.05, 0.5, 0.6, 0.95$ for $(x_1, x_2) = (0,0), (0,1), (1,0), (1,1)$; $P(x_3'{=}1) = 0.8$ if $x_3 = 1$, else 0.2. From 3000 simulated transitions (seed 2026), BIC-score, for each next-state factor, all $1 + 4 + 6 = 11$ candidate parent sets of size ≤ 2 drawn from $\{x_1, x_2, x_3, a\}$.
Solution (simulation output). The winners are the true parent sets: $\{x_1, a\}$ (BIC -1191.0 vs -1501.7 for the runner-up $\{a\}$), $\{x_1, x_2\}$ (-1364.3 vs -1715.3 for $\{x_1\}$), $\{x_3\}$ (-1504.8 vs -1511.1 for $\{x_1, x_3\}$). The agent learns that a only affects x_1; such sparse factored world models need far fewer samples than a flat transition table (model-based RL, course session 20).

PYTHON LAB · `s11_examples.py` · `ex_dbn_structure()`

```
x1':  best {x1, a} (-1191.0), runner-up {a} (-1501.7); x2':  {x1, x2}
(-1364.3), runner-up {x1} (-1715.3)
x3':  best {x3} (-1504.8), runner-up {x1, x3} (-1511.1)
```

AI CONNECTION · Multi-Agent AI

Five agents log continuous action signals (simulation, seed 2026, $n = 2000$ hypothetical rounds of the linear-Gaussian model $A_1 = \varepsilon_1$, $A_2 = 0.8A_1 + 0.6\varepsilon_2$, $A_3 = 0.7A_2 + 0.7\varepsilon_3$, $A_4 = 0.6A_1 + 0.8\varepsilon_4$, $A_5 = \varepsilon_5$ with independent standard normal ε_j): A_1 influences A_2 and A_4, A_2 influences A_3, A_5 acts alone. Marginal correlations link almost everyone: $\mathrm{corr}(A_3, A_4) = 0.35$. Partial correlations given the other three agents (Session 5: zeros of the precision matrix) with Fisher-z tests at level 0.01 ($|S| = 3$, so $z = \frac{1}{2}\ln\frac{1+\hat{r}}{1-\hat{r}}\sqrt{2000-6}$) recover exactly the interaction graph $A_1 - A_2$, $A_1 - A_4$, $A_2 - A_3$; the partial correlation of A_3, A_4 is 0.00 (Fig. 11.9, right). As in Worked Example 11.4, the graph is undirected where observational data cannot tell leaders from followers (`ex_multiagent_graph()`).

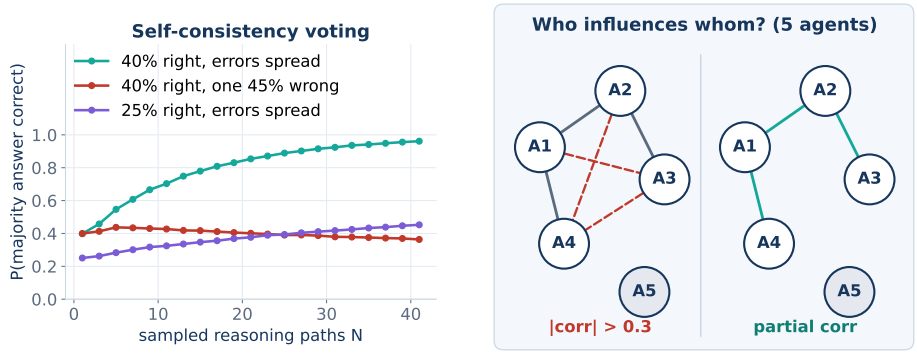


Figure 11.9. Left: self-consistency voting over N sampled reasoning paths (hypothetical answer distributions, 60,000 simulated questions per point): the two scenarios of the Agentic AI box, plus a weaker model that is right 25% of the time with four wrong answers at 18.75% each — the right answer is still the most likely one, so accuracy still rises, slowly. Right: thresholding marginal correlations at $|\mathrm{corr}| > 0.3$ invents interactions (red dashed); partial correlations recover the true interaction graph of five agents.

LINK TO COURSE SESSION 27 · Multi-Agent AI

Learning who coordinates with whom is structure learning; the learned graph becomes the coordination graph for max-sum or message passing.

PART V · AI CONNECTIONS

11.19 Summary

THINK ABOUT IT **(1) Two DAGs differing in one edge direction get the same BIC. What does that say about using a learned structure to predict interventions? (2) All chains report $\hat{R} = 1.00$, yet no trace ever switches between two known modes. Converged? (3) When should a real-time agent prefer biased-but-fast VI to asymptotically exact MCMC?**

Table 11.3. This session in the Advanced AI course (plain numbers are course sessions; "Session k" refers to these notes).

Concept today	Where it reappears	Role there
Monte Carlo averages, CLT	RL and MDPs (17–18), MCTS (22)	policy evaluation, rollouts, UCT bonus
Importance weights, ESS	off-policy RL (18, 20), causal effects (Session 15)	per-decision IS, inverse propensity weighting
Particle filtering	decisions under uncertainty (21), agents (25)	belief tracking, localisation
MCMC, VI, ELBO	generative models (4), learning PGMs (Session 10)	energy-based models, VAEs, diffusion, EM
Structure learning, CPDAGs	causal graphs and faithfulness (Sessions 12–13), MARL (27), world models (20)	what data can and cannot orient
Self-consistency sampling	agentic AI (26), frontiers (31)	test-time compute, verification

Table 11.4. Cheat sheet of key formulas.

Item	Formula
Likelihood of a DAG	$\ell = N\sum_i \hat{I}(X_i;U_i) - N\sum_i \hat{H}(X_i)$; AIC $= \ell - d$, BIC $= \ell - \frac{d}{2}\log N$
Bayesian score	$\prod_{i,u} \frac{\Gamma(\alpha_u)}{\Gamma(\alpha_u+N_u)} \prod_x \frac{\Gamma(\alpha_{x\mid u}+N_{x,u})}{\Gamma(\alpha_{x\mid u})}$
Monte Carlo	$\hat{\mu} = \frac{1}{N}\sum f(x^{(m)})$, SE $= \sigma/\sqrt{N}$; CI $\hat{\mu} \pm 1.96\, s/\sqrt{N}$
Importance sampling	$w = p/q$; $q^* \propto \lvert f\rvert p$; ESS $= (\sum w)^2/\sum w^2$
Metropolis–Hastings	$\alpha = \min\{1, \tilde{\pi}(x')q(x \mid x')/[\tilde{\pi}(x)q(x' \mid x)]\}$
MCMC error	ESS $= N/\tau$, $\tau = 1 + 2\sum_k \rho_k$; $\hat{R} = \sqrt{((n-1)W/n + B/n)/W}$
VI	$\log p(x) = \mathrm{ELBO}(q) + \mathrm{KL}(q\Vert p(\cdot \mid x))$; $\log q_j^* = \mathbb{E}_{-j} \log p(x,z) + c$

Table 11.5. Common pitfalls and how to avoid them.

Pitfall	Remedy
Structure by likelihood alone	penalise: BIC, BDe, held-out likelihood
Reading a CPDAG causally	needs faithfulness, no hidden confounders (Sessions 12–13)
Reporting N, not the ESS	ESS $= (\sum w)^2/\sum w^2$ or N/τ; heavy-tailed q
High acceptance = good mixing	tune the step; several chains, ACF, $\hat{R}$
Trusting VI uncertainty	mean field shrinks variances (Proposition 11.8)

Further reading. Koller & Friedman (2009), Chs. 12 (sampling), 11 (variational), 18 (structure learning); Bishop (2006), Chs. 10–11; Murphy (2023), Chs. 10–13; Spirtes, Glymour & Scheines (2000); Robert & Casella (2004); Wasserman (2004), Ch. 24; Blei, Kucukelbir & McAuliffe (2017).

APPENDIX

11.20 Appendix: DAG Counting and Detailed Balance

Robinson's recurrence. The number $a(n)$ of DAGs on n labeled nodes satisfies $a(0) = 1$ and

$$a(n) = \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} 2^{k(n-k)} a(n-k),$$

obtained by an inclusion-exclusion argument over the possible sets of "source" nodes (nodes with no incoming edges): choose k nodes to be sources ($\binom{n}{k}$ ways), let each of the $k(n-k)$ possible edges from a source to a non-source be present or absent ($2^{k(n-k)}$ ways), and put any DAG on the remaining $n-k$ nodes ($a(n-k)$ ways); every DAG has at least one source, and the alternating signs correct for DAGs counted with several source sets. Check: $a(2) = \binom{2}{1}2^1 a(1) - \binom{2}{2}2^0 a(0) = 4 - 1 = 3$ and $a(3) = 3 \cdot 2^2 \cdot 3 - 3 \cdot 2^2 \cdot 1 + 1 = 36 - 12 + 1 = 25$.
Detailed balance for Gibbs sampling. Let $T(\mathbf{x} \to \mathbf{x}')$ be the Gibbs transition probability for resampling coordinate i (all other coordinates unchanged, $x_{-i} = x'_{-i}$). Detailed balance requires $P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(\mathbf{x}')T(\mathbf{x}' \to \mathbf{x})$. Since $T(\mathbf{x} \to \mathbf{x}') = P(x'_i \mid x_{-i})$ by construction, and using $P(\mathbf{x}) = P(x_i \mid x_{-i})P(x_{-i})$,

$$P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(x_i \mid x_{-i})P(x_{-i})\, P(x'_i \mid x_{-i}) = P(x_{-i})P(x_i \mid x_{-i})P(x'_i \mid x_{-i}),$$

which is symmetric in $x_i \leftrightarrow x'_i$ (using $x_{-i} = x'_{-i}$), so it equals $P(\mathbf{x}')T(\mathbf{x}' \to \mathbf{x})$ by the same computation with roles swapped. Detailed balance implies P is a stationary distribution of this single-coordinate update (summing the balance equation over $\mathbf{x}$ gives $\sum_{\mathbf{x}} P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(\mathbf{x}')$, the definition of stationarity). A full sweep applies the updates $T_1, \dots, T_n$ in turn; if $\mathbf{x} \sim P$ before T_i then $\mathbf{x} \sim P$ after it, so P is also stationary for the sweep $T_1 T_2 \cdots T_n$ (which is in general not itself reversible); ergodicity (needed for the chain to actually *converge* to this stationary distribution regardless of start) requires positivity of all full conditionals, which holds automatically whenever the Gibbs potentials are everywhere strictly positive.

CONNECTIONS AND TAKEAWAYS

We now have the complete toolkit of the probabilistic half of these notes: represent independence with a graph (Sessions 6, 8), compute exactly when structure allows it (Session 7), learn parameters and handle latent variables (Session 10), learn the graph and fall back to sampling — including MCMC — whenever exact computation is out of reach (today). Every one of these tools computes properties of a single, fixed joint distribution P obtained by *observing* the world.
Here is the question none of Sessions 5–11 can answer: if I *change* the world — give every patient the drug rather than merely observing who happened to take it, force an ad in front of every customer rather than observing who happened to see it — what happens to Y? This is not a question about $P(Y \mid X = x)$, and, as **Session 12** will demonstrate with real numbers, answering it as if it were can be actively wrong, sometimes even reversing the sign

of the true effect. That demonstration is the hinge of the entire course.